

The Construction of Near-Optimal Universal Coding of Integers

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Abstract—The Universal Coding of Integers (UCI) is suitable for discrete memoryless sources with unknown probability distributions and infinitely countable alphabet sizes. A UCI is a class of prefix codes for which the ratio of the average codeword length to $\max\{1, H(P)\}$ is within a constant expansion factor C_C for any decreasing probability distribution P , where $H(P)$ is the entropy of P . For any UCI code C , the minimum expansion factor C_C^* is defined to represent the infimum of the set of extension factors of C . Each C has a unique corresponding C_C^* , and the smaller C_C^* is, the better the compression performance of C is. The class of UCIs C (or a family $\{C_i\}_{i=1}^\infty$) that achieves the smallest C_C^* is defined as the *optimal UCI*. The best current result is that the range of C_C^* for the optimal UCI is $2 \leq C_C^* \leq 2.5$. In this paper, we prove a tighter probability inequality for decreasing distributions, which serves as a new tool for studying the properties of UCIs. On the basis of this inequality, we prove that there exists a class of near-optimal UCIs, called the ν code, achieving $C_\nu = 2.0386$. This narrows the range of the minimum expansion factor for the optimal UCI to $2 \leq C_C^* \leq 2.0386$. We show that the ν code is currently optimal in terms of the minimum expansion factor. In addition, we propose a new proof showing that the minimum expansion factor of the optimal UCI is lower bounded by 2.

Index Terms—Elias coding, source coding, minimum expansion factor, variable-length codes, universal coding of integers.

I. INTRODUCTION

In their pioneering work [1], [2] Shannon proposed a model for general communication systems, defined Shannon's entropy, initiated the study of source coding, and constructed the first variable-length code (VLC), which was termed the Shannon–Fano coding. Source coding improves the effectiveness of digital information storage, processing, and transmission methods by reducing redundancy. VLCs form an essential source coding class that encodes individual source symbols into variable-length codewords. The idea behind constructing a VLC is to map source symbols with higher occurrence probabilities into shorter codewords, thereby achieving a shorter average codeword length. This idea emerged as early as the nineteenth century, when Morse developed Morse code. When the probability distribution of discrete memoryless sources (DMSs) is known, the Huffman code [3] is the prefix

code with the shortest average codeword length, where the prefix code is a special type of VLC that has any codeword that is not a prefix of any other codeword. The arithmetic code [4], [5] is another well-known source code; however, it is a string coding method that falls outside the scope of this work.

However, in practical applications, it is sometimes difficult to know precisely about the statistical properties of the source or even to know anything. When the probability distribution of DMSs is unknown, the Universal Coding of Integers (UCI) is the best-known class of VLCs. A systematic study of UCIs was initiated by Elias [6] in 1975, and UCIs were determined to apply to DMSs of the countably infinite alphabet, with the assumption of no prior knowledge. Today, UCIs are widely used in various areas, including data deduplication schemes [7], [8], stream processing engines [9], [10], biological sequencing with data compression [11], [12], quantized stochastic gradient descent [13], and evolving secret sharing [14], [15].

Elias [6] considered the following universal coding problem. The source $S = (\mathcal{A}, P)$ is discrete and memoryless, where the alphabet $\mathcal{A} \triangleq \mathbb{N} = \{1, 2, \dots\}$ is countably infinite and the probability distribution P is decreasing (i.e., $P(a) \geq P(a+1)$ for all $a \in \mathcal{A}$ and $\sum_{k=1}^\infty P(k) = 1$).¹ Let $H(P) \triangleq -\sum_{k=1}^\infty P(k) \log_2 P(k)$ be the Shannon entropy of P . Let C be a prefix code for the DMS $S = (\mathcal{A}, P)$; that is, C maps the positive integers onto binary codewords and satisfies the prefix property. Let $L_C(\cdot)$ be the length function such that $L_C(a) = |\mathcal{C}(a)|$ for all $a \in \mathcal{A}$, where $\mathcal{C}(a)$ denotes the a -th codeword of C . Let $A_P(L_C) \triangleq \sum_{k=1}^\infty P(k) L_C(k)$ denote the average codeword length of C under the probability distribution P . The prefix code C is said to be *universal* if a constant C_C exists such that

$$\frac{A_P(L_C)}{\max\{1, H(P)\}} \leq C_C \quad (1)$$

for all P with $0 < H(P) < \infty$. The constant C_C is called the *expansion factor* of UCI C , and we let S_C denote the set consisting of all expansion factors of UCI C .

Since Elias's pioneering work [6], many UCIs have been proposed, and they can be categorized into two main groups [16]: *the message length scheme* and *the flag scheme*.

- 1) In the message length scheme, the process of encoding an integer $n \in \mathbb{N}$ can be divided into two parts, where the suffix part encodes n with a length of m bits and the prefix part encodes m . Note that the prefix part can be further encoded into two parts, and so forth. The γ ,

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¹In this paper, we assume that any probability distribution P is decreasing.

δ , and ω codes proposed by Elias [6] are typical UCIs within the message length scheme. A UCI $\mathcal{C}$ in this scheme mainly involves keeping the codeword length $L_{\mathcal{C}}(n)$ as small as possible for sufficiently large values of $n \in \mathbb{N}$. When the length of the codeword at infinity is as small as possible, the length of the codeword at smaller integers is relatively large; thus, this research focuses on the case where the Shannon entropy is large. UCIs for such schemes were proposed in [6], [17]–[20].

- 2) In the flag scheme, a specific string is used as a flag to identify the end of a codeword. For example, the Fibonacci code [21] is a typical UCI within the flag pattern scheme, and 11 is its flag. The UCIs in the message length scheme can catastrophically propagate errors into subsequent codewords during the decoding process. However, the flag scheme used by UCIs has a decoding synchronization property that prevents the infinite propagation of errors. UCIs for such schemes were proposed in [21]–[25].

Recently, Allison *et al.* [26] analysed the main properties of Wallace tree codes and proved that they are UCIs. [27], [28] defined the *minimum expansion factor* $C_{\mathcal{C}}^* \triangleq \inf S_{\mathcal{C}}$. For any UCI $\mathcal{C}$, its minimum expansion factor $C_{\mathcal{C}}^*$ is unique. The minimum expansion factor characterizes the compression performance of a UCI. If a class of UCIs $\mathcal{C}$ or a family of UCIs $\{\mathcal{C}_i\}_{i=1}^{\infty}$ possesses the smallest minimum expansion factor,² then $\mathcal{C}$ or $\{\mathcal{C}_i\}_{i=1}^{\infty}$ is said to be the *optimal UCI*. Yan *et al.* proved that the range of $C_{\mathcal{C}}^*$ for the optimal UCI is $2 \leq C_{\mathcal{C}}^* \leq 2.5$. In addition, a generalized universal coding of integers related to UCIs has been proposed and investigated [29]–[31].

In this paper, we further narrow the range of $C_{\mathcal{C}}^*$ for the optimal UCI. By establishing a tighter probability inequality for decreasing distributions and developing an improved research approach, we prove that the constructed ν code is currently optimal in terms of the minimum expansion factor. The contributions of this paper are listed below.

- 1) Inspired by minimax regret data compression, we propose a new proof showing that the minimum expansion factor of the optimal UCI is lower bounded by 2.
- 2) We prove a tighter probability inequality for decreasing distributions, which serves as a new tool for studying the properties of UCIs. On the basis of this inequality, we improve the research approach for determining the upper bound of the minimum expansion factor for the optimal UCI.
- 3) We propose a new class of UCIs, termed the ν code, and prove that the ν code is currently optimal in terms of the minimum expansion factor with $C_{\nu} = 2.0386$. This reduces the upper bound of the minimum expansion factor for the optimal UCI from 2.5 to 2.0386.

The paper is structured as follows. Section II provides the relevant conclusions regarding UCIs and Abel's formula. In Section III, a new proof stating that the minimum expansion

factor of the optimal UCI is lower bounded by 2 is proposed. Section IV presents a tighter probability inequality for decreasing distributions and introduces an improved research approach. Section V involves the construction of a new class of UCIs, termed the ν code. In Section VI, the upper bound of C_{ν}^* is proven. Section VII provides the lower bound of C_{ν}^* and briefly compares the range of C_{ν}^* with that of $C_{\mathcal{C}}^*$ for the classic UCIs. Section VIII concludes this work.

II. PRELIMINARIES

In this section, we present the Elias δ code, recent advances regarding minimum expansion factors for UCIs and Abel's formula. Table I presents the main notations used in this paper.

A. The Elias δ code

Let $\{0, 1\}^*$ denote all binary strings of finite length and $|\xi|$ denote the length of string ξ . Let an α code be a unary encoding [6], [32]. The unary code of $a \in \mathcal{A}$ consists of $a - 1$ zeros followed by a single one. For example, $\alpha(1) = 1$, and $\alpha(4) = 0001$. Let a β code be a binary representation. For example, $\beta(4) = 100$, and $\beta(11) = 1011$. Let $[\beta(a)]$ denote the removal of the most significant bit 1 of $\beta(a)$. In particular, $[\beta(1)]$ is a null string. Then, the codeword length is as follows.

$$\begin{aligned} |\alpha(a)| &= a, \\ |\beta(a)| &= 1 + \lfloor \log_2 a \rfloor, \\ |[\beta(a)]| &= \lfloor \log_2 a \rfloor \end{aligned}$$

for all $a \in \mathcal{A}$. The Elias γ code [6]: $\mathcal{A} \rightarrow \{0, 1\}^*$ can be succinctly represented as

$$\gamma(a) = \alpha(|\beta(a)|)[\beta(a)]$$

for all $a \in \mathcal{A}$. The Elias δ code [6]: $\mathcal{A} \rightarrow \{0, 1\}^*$ can be succinctly represented as

$$\delta(a) = \gamma(|\beta(a)|)[\beta(a)]$$

for all $a \in \mathcal{A}$. Afterward, we obtain

$$\begin{aligned} L_{\delta}(a) &= |\gamma(|\beta(a)|)| + |[\beta(a)]| \\ &= |\gamma(1 + \lfloor \log_2 a \rfloor)| + \lfloor \log_2 a \rfloor \\ &= |\alpha(|\beta(1 + \lfloor \log_2 a \rfloor)|)| + |[\beta(1 + \lfloor \log_2 a \rfloor)]| + \lfloor \log_2 a \rfloor \\ &= |\beta(1 + \lfloor \log_2 a \rfloor)| + \lfloor \log_2(1 + \lfloor \log_2 a \rfloor) \rfloor + \lfloor \log_2 a \rfloor \\ &= 1 + \lfloor \log_2 a \rfloor + 2\lfloor \log_2(1 + \lfloor \log_2 a \rfloor) \rfloor \end{aligned}$$

for all $a \in \mathcal{A}$. Table II lists the first 16 codewords of the Elias γ and Elias δ codes. Additional details related to the Elias code can be found at [6].

B. The minimum expansion factor of UCIs

The minimum expansion factor $C_{\mathcal{C}}^*$ of a UCI $\mathcal{C}$ characterizes the compression performance of $\mathcal{C}$; that is, whatever the probability distribution P of the source is, the average codeword length $A_P(L_{\mathcal{C}})$ is less than or equal to $C_{\mathcal{C}}^*$ times $\max\{1, H(P)\}$. Elias [6] proved that the minimum expansion factor $C_{\mathcal{C}}^*$ has a trivial lower bound of 1, and Yan *et al.* [27]

²The minimum expansion factor corresponding to a family of UCIs $\{\mathcal{C}_i\}_{i=1}^{\infty}$ is $\lim_{i \rightarrow \infty} C_{\mathcal{C}_i}^*$.

TABLE I
MAIN NOTATIONS USED IN THIS PAPER

Notation	Description	Definition
$\mathcal{A}$	The countably infinite alphabet	Section I
P	The decreasing probability distribution	Section I
$S = (\mathcal{A}, P)$	The discrete memoryless source with an alphabet $\mathcal{A}$ and a probability distribution P	Section I
$H(P)$	The Shannon entropy of P	Section I
$\mathcal{C}$	The prefix code for $S = (\mathcal{A}, P)$	Section I
$\mathcal{C}(a)$	The a -th codeword of $\mathcal{C}$ for $a \in \mathcal{A}$	Section I
$L_{\mathcal{C}}(a)$	The length of $\mathcal{C}(a)$	Section I
$A_P(L_{\mathcal{C}})$	The average codeword length of $\mathcal{C}$ under a probability distribution P	Section I
$C_{\mathcal{C}}$	The expansion factor of UCI $\mathcal{C}$	Eq. (1)
$S_{\mathcal{C}}$	The set consisting of all expansion factors of UCI $\mathcal{C}$	Section I
$C_{\mathcal{C}}^*$	The minimum expansion factor of UCI $\mathcal{C}$	Section I
$\{0, 1\}^*$	The set consisting of all binary strings of finite length	Section II
$ \xi $	The length of string ξ	Section II
$\mathcal{P}_{\mathcal{A}}$	The set of all decreasing probability distributions over $\mathcal{A}$ with finite Shannon entropy	Section III
$\mathcal{UCI}$	The set of all UCIs	Section III
C^*	The minimum expansion factor of the optimal UCI	Section III
$h(p)$	The binary entropy of $(p, 1 - p)$	Section III
A_n	$A_n = \log_2 P(n) - \log_2(1 - P(1)) + \log_2 n$	Eq. (10)
C_n	$C_n = (1 + \frac{1}{n})(1 - \frac{1}{n})^{n-1}$	Section IV
$g_{(c_1, c_2)}(a, P(1))$	$g_{(c_1, c_2)}(a, P(1)) = \frac{c_1}{c_2 - \log_2(1 - P(1))} (\log_2 a - \log_2(1 - P(1)))$	Eq. (25)
$h_{(c_1, c_2)}(t, x)$	$h_{(c_1, c_2)}(t, x) = \left(\frac{c_1}{c_2 + x} - 1\right)t + \frac{c_1 x}{c_2 + x} - 1$	Eq. (28)
$D_{(c_1, c_2)}(P(1))$	$D_{(c_1, c_2)}(P(1)) = \frac{c_1}{c_2 - \log_2(1 - P(1))}$	Eq. (26)
$J_{(c_1, c_2)}(P(1), L)$	$J_{(c_1, c_2)}(P(1), L) = D_{(c_1, c_2)}(P(1)) \log_2 C_L + P(1) \left[1 + D_{(c_1, c_2)}(P(1)) (\log_2 P(1) - \log_2 C_L)\right]$	
$R_{(c_1, c_2)}(P(1), L)$	$R_{(c_1, c_2)}(P(1), L) = 3 + D_{(c_1, c_2)}(P(1)) (\log_2(1 - P(1)) + \sum_{n=2}^{L-1} \log_2 C_n - (L-1) \log_2 C_L)$	
$Q_{(c_1, c_2)}(P(1), P(2), L)$	$Q_{(c_1, c_2)}(P(1), P(2), L) = J_{(c_1, c_2)}(P(1), L) + P(2) R_{(c_1, c_2)}(P(1), L)$	

TABLE II
THE FIRST 16 CODEWORDS OF THE ELIAS γ AND ELIAS δ CODES

n	Elias γ code	Elias δ code
1	1	1
2	01 0	010 0
3	01 1	010 1
4	001 00	011 00
5	001 01	011 01
6	001 10	011 10
7	001 11	011 11
8	0001 000	00100 000
9	0001 001	00100 001
10	0001 010	00100 010
11	0001 011	00100 011
12	0001 100	00100 100
13	0001 101	00100 101
14	0001 110	00100 110
15	0001 111	00100 111
16	00001 0000	00101 0000

TABLE III
THE MINIMUM EXPANSION FACTORS $C_{\mathcal{C}}^*$ OF SOME UCIs

Code	The range of $C_{\mathcal{C}}^*$
γ code [6]	$C_{\gamma}^* = \mathbf{3}$
δ code [6]	$2.5 \leq C_{\delta}^* \leq \mathbf{2.75}$
ω code [6]	$2.1 < C_{\omega}^* \leq \mathbf{3}$
$CE(k)$ code [33]	$2 + k \leq C_{CE(k)}^* \leq \mathbf{2 + k + \frac{1}{2k}}$
η code [27]	$2.5 \leq C_{\eta}^* \leq \frac{\mathbf{8}}{\mathbf{3}}$
θ code [27]	$2.5 \leq C_{\theta}^* \leq \mathbf{2.8}$
ι code [28]	$C_{\iota}^* = \mathbf{2.5}$
κ code [28]	$2.5 \leq C_{\kappa}^* \leq \frac{\mathbf{8}}{\mathbf{3}}$
$\kappa[t]$ code [28]	$2.5 \leq C_{\kappa[t]}^* \leq \mathbf{2.5 + \frac{1}{2t+2}}$

$$L_{\mathcal{C}}(a) \geq 1 + \lfloor \log_2 a \rfloor \text{ for all } a \in \mathcal{A}.$$

Table III [28] summarizes the latest research results that have been published regarding the minimum expansion factors $C_{\mathcal{C}}^*$ of some UCIs. In Table III, the ι code [28] constructed by Yan *et al.* can reach an expansion factor $C_{\iota} = 2.5$. Thus, the range of $C_{\mathcal{C}}^*$ for the optimal UCI is $2 \leq C_{\mathcal{C}}^* \leq 2.5$.

C. Abel's formula

We introduce Abel's formula, which is used in the proof of this paper.

Theorem 2 (Abel's formula). [34] Suppose that $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are two sets of real numbers. Let $z_k \triangleq b_1 + b_2 + \dots + b_k$ for $k = 1, 2, \dots, n$; then,

$$\sum_{i=1}^n a_i b_i = (a_1 - a_2) z_1 + (a_2 - a_3) z_2 + \dots + (a_{n-1} - a_n) z_{n-1} + a_n z_n.$$

first introduced a nontrivial lower bound for $C_{\mathcal{C}}^*$, as described below.

Theorem 1. [27] The minimum expansion factor $C_{\mathcal{C}}^* \geq 2$ for any UCI $\mathcal{C}$.

Theorem 1 was proven in [27] by the following lemma.

Lemma 1. [27] Suppose that $\mathcal{C}$ is any given UCI and that K_1 and K_2 are two positive constants.

- 1) If the length function $L_{\mathcal{C}}(\cdot)$ satisfies $L_{\mathcal{C}}(a) \geq K_1 + K_2 \lfloor \log_2 a \rfloor$ for all $a \in \mathcal{A}$, then $C_{\mathcal{C}}^* \geq K_1 + K_2$.
- 2) If $L_{\mathcal{C}}(\cdot)$ satisfies $L_{\mathcal{C}}(a) \leq L_{\mathcal{C}}(a+1)$ for all $a \in \mathcal{A}$, then

Proof. We obtain

$$\begin{aligned} & (a_1 - a_2)z_1 + (a_2 - a_3)z_2 + \cdots + (a_{n-1} - a_n)z_{n-1} + a_n z_n \\ &= a_1 z_1 + a_2(z_2 - z_1) + \cdots + a_n(z_n - z_{n-1}) \\ &= a_1 b_1 + a_2 b_2 + \cdots + a_n b_n. \end{aligned}$$

□

III. THE MINIMUM EXPANSION FACTOR OF THE OPTIMAL UCI

This section revisits the minimal expansion factor of the optimal UCI and provides a new proof of Theorem 1. The work in this section is inspired by minimax regret data compression [35, chap.13].

Let $\mathcal{P}_A$ be the set of all decreasing probability distributions over a countably infinite alphabet $\mathcal{A}$ with finite Shannon entropy values. Let $\mathcal{UCI}$ denote the set of all UCIs. For any UCI $\mathcal{C}$, the minimum expansion factor $C_{\mathcal{C}}^*$ can be equivalently defined as follows:

$$\begin{aligned} C_{\mathcal{C}}^* &= \inf S_{\mathcal{C}} \\ &= \inf \left\{ C_{\mathcal{C}} \mid \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}} \leq C_{\mathcal{C}} \text{ for } \forall P \in \mathcal{P}_A \right\} \\ &= \sup_{P \in \mathcal{P}_A} \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}}. \end{aligned}$$

Furthermore, the minimum expansion factor of the optimal UCI can be defined as follows:

$$\begin{aligned} C^* &= \inf_{\mathcal{C} \in \mathcal{UCI}} C_{\mathcal{C}}^* \\ &= \inf_{\mathcal{C} \in \mathcal{UCI}} \sup_{P \in \mathcal{P}_A} \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}}. \end{aligned} \quad (2)$$

Thus, we can precisely describe the problem of finding an optimal UCI as finding the exact value of C^* and constructing a UCI $\mathcal{C}$ or a family of UCIs $\{\mathcal{C}_i\}_{i=1}^{\infty}$ achieving C^* . The work of [27], [28] can be described as having determined that the range of C^* is $2 \leq C^* \leq 2.5$. Next, we provide a new proof of Theorem 1.

Proof. According to Equation (2), Theorem 1 is equivalent to $C^* \geq 2$. We prove that $C^* \geq 2$ as follows.

First, we construct some probability distribution P_m , where m is an integer greater than 1.

$$P_m(a) = \begin{cases} 1 - \frac{1}{m}, & \text{if } a = 1, \\ \frac{1}{m \cdot 2^m}, & \text{if } a = 2, 3, \dots, 2^m + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Letting $\mathcal{P}_A(M) \triangleq \{P_m \mid 2 \leq m \in \mathbb{N}\}$, it is clear that $\mathcal{P}_A(M) \subsetneq \mathcal{P}_A$. For all $2 \leq m \in \mathbb{N}$, we obtain

$$\begin{aligned} H(P_m) &= -P(1) \log_2 P(1) - \sum_{a=2}^{2^m+1} P_m(a) \log_2 P_m(a) \\ &= -\left(1 - \frac{1}{m}\right) \log_2 \left(1 - \frac{1}{m}\right) - \frac{2^m}{m 2^m} \log_2 \frac{1}{m 2^m} \\ &= 1 + h\left(\frac{1}{m}\right), \end{aligned}$$

where $h(p) \triangleq -p \log_2 p - (1-p) \log_2 (1-p)$ denotes the binary entropy of $(p, 1-p)$.

For all $\mathcal{C} \in \mathcal{UCI}$ and all $\tilde{P} \in \mathcal{P}_A$,

$$\sup_{P \in \mathcal{P}_A} \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}} \geq \frac{A_{\tilde{P}}(L_{\mathcal{C}})}{\max\{1, H(\tilde{P})\}}$$

by definition. Hence, we have that

$$\inf_{\mathcal{C} \in \mathcal{UCI}} \sup_{P \in \mathcal{P}_A} \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}} \geq \inf_{\mathcal{C} \in \mathcal{UCI}} \frac{A_{\tilde{P}}(L_{\mathcal{C}})}{\max\{1, H(\tilde{P})\}} \quad (3)$$

for all $\tilde{P} \in \mathcal{P}_A$. Based on Equation (3) and the fact that $\mathcal{P}_A(M) \subsetneq \mathcal{P}_A$, we have

$$\begin{aligned} C^* &= \inf_{\mathcal{C} \in \mathcal{UCI}} \sup_{P \in \mathcal{P}_A} \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}} \\ &\geq \sup_{P \in \mathcal{P}_A} \inf_{\mathcal{C} \in \mathcal{UCI}} \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}} \\ &\geq \sup_{P \in \mathcal{P}_A(M)} \inf_{\mathcal{C} \in \mathcal{UCI}} \frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}} \\ &\geq \inf_{\mathcal{C} \in \mathcal{UCI}} \frac{A_{P_m}(L_{\mathcal{C}})}{\max\{1, H(P_m)\}} \end{aligned} \quad (4)$$

for all $2 \leq m \in \mathbb{N}$. When the probability distribution P_m is given,³ the Huffman code [3] is optimal; i.e., the average codeword length of the Huffman code is the shortest. Therefore, we obtain

$$\begin{aligned} C^* &\geq \inf_{\mathcal{C} \in \mathcal{UCI}} \frac{A_{P_m}(L_{\mathcal{C}})}{\max\{1, H(P_m)\}} \\ &= \frac{1}{1 + h(\frac{1}{m})} \inf_{\mathcal{C} \in \mathcal{UCI}} A_{P_m}(L_{\mathcal{C}}) \\ &\stackrel{(b)}{>} \frac{1}{1 + h(\frac{1}{m})} \left[1 \times \left(1 - \frac{1}{m}\right) + 2^m \times (m+1) \times \frac{1}{m \cdot 2^m} \right] \\ &= \frac{2}{1 + h(\frac{1}{m})} \end{aligned} \quad (5)$$

for all $2 \leq m \in \mathbb{N}$, where (b) is true because $\mathcal{C}$ is a UCI, which cannot consider only the first $2^m + 1$ codewords, and to make Kraft's inequality [35], [36] hold, the average codeword length of the first $2^m + 1$ codewords of a UCI code must therefore be strictly longer than that of a Huffman code. From Equation (5), we have

$$\begin{aligned} C^* &\geq \sup_{2 \leq m \in \mathbb{N}} \frac{2}{1 + h(\frac{1}{m})} \\ &= \lim_{m \rightarrow \infty} \frac{2}{1 + h(\frac{1}{m})} \\ &= 2. \end{aligned} \quad (6)$$

□

IV. RESEARCH APPROACHES FOR DETERMINING THE UPPER BOUND OF C^*

In this section, we review the research approaches that have been used to address the upper bound of C^* and provide inequalities that are tighter than those listed in previous work. On the basis of these inequalities, we propose an improved research approach.

³For any fixed m , the probability distribution P_m is defined over a finite alphabet $\{1, 2, \dots, 2^m + 1\}$.

A. Review of research on the upper bound of C^*

The main work that has been conducted on the upper bound of C^* is as follows. In 1975, the expansion factor of the γ code constructed by Elias [6] was determined as $C_\gamma = 3$, thus obtaining the first upper bound 3 for C^* . Recently, the η [27] and ι [28] codes constructed by Yan and Lin could achieve expansion factors $C_\eta = 2.75$ and $C_\iota = 2.5$, respectively; thus, the upper bound of C^* was compressed to 2.5. The research approaches employed in these three studies were consistent, as detailed below.

First, for a given UCI $\mathcal{C}$, it must be proven the codeword length satisfies $L_{\mathcal{C}}(1) = 1$ and $L_{\mathcal{C}}(a) \leq K_1 + K_2 \log_2 a$ for all $2 \leq a \in \mathcal{A}$, where K_1 and K_2 are two positive constants. Consequently, we obtain

$$\begin{aligned} A_P(L_{\mathcal{C}}) &\leq P(1) + \sum_{a=2}^{\infty} P(a)(K_1 + K_2 \log_2 a) \\ &= K_1 + (1 - K_1)P(1) + K_2 \sum_{a=2}^{\infty} P(a) \log_2 a. \end{aligned}$$

Second, we need to use an inequality to obtain an upper bound for the expected value of $\log_2 a$. Elias [6] used Wyner's inequality [37]:

$$\sum_{a=1}^{\infty} P(a) \log_2 a = \sum_{a=2}^{\infty} P(a) \log_2 a \leq H(P), \quad (8)$$

and Yan *et al.* [28] used

$$\sum_{a=2}^{\infty} P(a) \log_2 a \leq H(P) + P(1) \log_2 P(1). \quad (9)$$

Next,

$$\frac{A_P(L_{\mathcal{C}})}{\max\{1, H(P)\}} \leq F_1(H(P), P(1))$$

is obtained through the above inequalities, where $F_1(H(P), P(1))$ is a function of $H(P)$ and $P(1)$. Finally, the upper bound of the function $F_1(H(P), P(1))$ is analysed. Equation (7) is often used in the analysis process, shown at the bottom of the page, simplifying $F_1(H(P), P(1))$ to $F_1(P(1))$; that is, F_1 is independent of $H(P)$.

In short, three key points are involved in proving the upper bound of C^* . First, an inequality for the upper bound of the codeword length is needed, such as $L_{\mathcal{C}}(a) \leq K_1 + K_2 \log_2 a$. We call such an inequality a *codeword length upper bound inequality*; such inequalities are related only to the codeword construction process and are independent of the underlying distribution. Second, an inequality for the upper bound of the expected value of $\log_2 a$ is needed, such as Inequalities (8) and (9). We call such inequalities *probability inequalities for decreasing distributions*. This type of inequality is the core of the proof. Third, the parameters of the upper bound function in the final analysis, for example, the parameters of F_1 , are $H(P)$ and $P(1)$.

B. The new probability inequality for decreasing distributions

In this subsection, we provide a new probability inequality for decreasing distributions and prove that this inequality is tighter than Inequalities (8) and (9);⁴ that is, it has a smaller upper bound.

Before the new inequality is given, we first define the relevant notations. Let $T_P \triangleq \sup\{a \in \mathcal{A} \mid P(a) > 0\}$ for any $P \in \mathcal{P}_{\mathcal{A}}$. Note that if $P(a) > 0$ for any $a \in \mathcal{A}$, then $T_P = +\infty$. Let

$$A_n \triangleq \log_2 P(n) - \log_2(1 - P(1)) + \log_2 n \quad (10)$$

for all $n \in \mathbb{N} \cap [2, T_P]$. When $T_P < +\infty$, A_{n_0} is undefined for $n_0 > T_P$. However, owing to $0 \log_2 0 \triangleq 0$,⁵ we can define

$$P(n_0)A_{n_0} = 0[\log_2 0 - \log_2(1 - P(1)) + \log_2 n_0] \triangleq 0,$$

for all $n_0 > T_P$.

Lemma 2. *The inequality*

$$\sum_{n=2}^{\infty} P(n)A_n \leq P(2) + \sum_{n=2}^{\infty} P(n+1) \log_2 \left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n}\right)^{n-1} \quad (11)$$

holds for any $P \in \mathcal{P}_{\mathcal{A}}$. Inequality (11) can be equivalently rewritten as follows:

$$\begin{aligned} \sum_{n=2}^{\infty} P(n) \log_2 n &\leq H(P) - h(P(1)) + P(2) \\ &\quad + \sum_{n=2}^{\infty} P(n+1) \log_2 \left(1 + \frac{1}{n}\right) \left(1 - \frac{1}{n}\right)^{n-1}. \end{aligned} \quad (12)$$

Proof. Let $B_n \triangleq \sum_{k=2}^n A_k$ for all $n \in \mathbb{N} \cap [2, T_P]$. From the arithmetic mean–geometric mean inequality, we obtain

$$\left(\prod_{k=2}^n P(k)\right)^{\frac{1}{n-1}} \leq \frac{\sum_{k=2}^n P(k)}{n-1} \leq \frac{1 - P(1)}{n-1}$$

for any integer $n \geq 2$. Furthermore, we have that

$$\begin{aligned} B_n &= \log_2 \prod_{k=2}^n P(k) - (n-1) \log_2(1 - P(1)) + \log_2 n! \\ &\leq \log_2 \left(\frac{1 - P(1)}{n-1}\right)^{n-1} - \log_2(1 - P(1))^{n-1} + \log_2 n! \\ &= \log_2 \frac{n!}{(n-1)^{n-1}}. \end{aligned}$$

⁴Because $H(P) + P(1) \log_2 P(1) \leq H(P)$, Inequality (9) is tighter than Inequality (8). Therefore, it is only necessary to show that the new inequality is tighter than Inequality (9).

⁵In mathematics, $0 \log_2 0$ can be defined as $0 \log_2 0 \triangleq \lim_{x \rightarrow 0^+} x \log_2 x = 0$.

$$\frac{K_3 + K_4 H(P)}{\max\{1, H(P)\}} = \begin{cases} K_3 + K_4 H(P) \leq K_3 + K_4, & \text{if } 0 \leq H(P) \leq 1, \\ \frac{K_3}{H(P)} + K_4 \leq K_3 + K_4, & \text{if } 1 \leq H(P) < \infty. \end{cases} \quad (7)$$

From Abel's formula, we obtain □

$$\begin{aligned}
& \sum_{n=2}^m P(n)A_n \\
&= (P(2) - P(3))B_2 + (P(3) - P(4))B_3 \\
&\quad + \cdots + (P(m-1) - P(m))B_{m-1} + P(m)B_m \\
&= P(m)B_m + \sum_{n=2}^{m-1} (P(n) - P(n+1))B_n \\
&\leq P(m) \log_2 \frac{m!}{(m-1)^{m-1}} \\
&\quad + \sum_{n=2}^{m-1} (P(n) - P(n+1)) \log_2 \frac{n!}{(n-1)^{n-1}} \\
&= P(2) + \sum_{n=2}^{m-1} P(n+1) \left(\log_2 \frac{(n+1)!}{n^n} - \log_2 \frac{n!}{(n-1)^{n-1}} \right) \\
&= P(2) + \sum_{n=2}^{m-1} P(n+1) \log_2 \left(\frac{(n+1)!}{n^n} \cdot \frac{(n-1)^{n-1}}{n!} \right) \\
&= P(2) + \sum_{n=2}^{m-1} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1}
\end{aligned} \tag{13}$$

for all $m \in \mathbb{N} \cap [2, T_P]$. We consider the following two cases.

1) When $T_P = +\infty$, owing to Equation (13),

$$\begin{aligned}
& \sum_{n=2}^{\infty} P(n)A_n \\
&= \lim_{m \rightarrow \infty} \sum_{n=2}^m P(n)A_n \\
&\leq \lim_{m \rightarrow \infty} \left[P(2) + \sum_{n=2}^{m-1} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} \right] \\
&= P(2) + \lim_{m \rightarrow \infty} \left[\sum_{n=2}^{m-1} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} \right] \\
&= P(2) + \sum_{n=2}^{\infty} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1}.
\end{aligned}$$

2) When $T_P < +\infty$, we consider Equation (13) by taking $m = T_P$; then,

$$\begin{aligned}
& \sum_{n=2}^{T_P} P(n)A_n \\
&\leq P(2) + \sum_{n=2}^{T_P-1} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1}.
\end{aligned}$$

Furthermore, we obtain

$$\begin{aligned}
& \sum_{n=2}^{\infty} P(n)A_n = 0 + \sum_{n=2}^{T_P} P(n)A_n \\
&\leq P(2) + \sum_{n=2}^{T_P-1} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} \\
&= P(2) + \sum_{n=2}^{\infty} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1}.
\end{aligned}$$

At the end of this subsection, we prove that Inequality (12) is tighter than Inequality (9). However, we first need to prove an auxiliary lemma. For convenience, let $C_n \triangleq (1 + \frac{1}{n}) (1 - \frac{1}{n})^{n-1}$ for all $2 \leq n \in \mathbb{N}$ in this paper.

Lemma 3. *The sequence $\{C_n\}_{n=2}^{\infty}$ is strictly monotonically decreasing.*

Proof. First, $1 + \frac{1}{n}$ is clearly strictly monotonically decreasing with respect to n . Second, the sequence

$$\left\{ \left(1 - \frac{1}{n} \right)^{n-1} = \left(\frac{n-1}{n} \right)^{n-1} \right\}_{n=2}^{\infty}$$

being strictly monotonically decreasing is equivalent to the sequence $\left\{ \left(\frac{n}{n-1} \right)^{n-1} \right\}_{n=2}^{\infty}$ being strictly monotonically increasing. From the arithmetic mean–geometric mean inequality, we have that

$$\begin{aligned}
\left(\frac{n}{n-1} \right)^{n-1} &= \underbrace{\frac{n}{n-1} \cdots \frac{n}{n-1}}_{n-1} \times 1 \\
&< \left[\frac{\frac{n}{n-1} \times (n-1) + 1}{n} \right]^n \\
&= \left(\frac{n+1}{n} \right)^n.
\end{aligned}$$

In conclusion, $\{C_n\}_{n=2}^{\infty}$ is strictly monotonically decreasing. □

Lemma 4. *The inequality*

$$\begin{aligned}
& H(P) - h(P(1)) + P(2) + \sum_{n=2}^{\infty} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} \\
&\leq H(P) + P(1) \log_2 P(1)
\end{aligned} \tag{14}$$

holds for any $P \in \mathcal{P}_{\mathcal{A}}$. Inequality (14) can be equivalently rewritten as follows:

$$\begin{aligned}
& (1 - P(1)) \log_2 (1 - P(1)) + P(2) \\
&+ \sum_{n=2}^{\infty} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} \leq 0.
\end{aligned} \tag{15}$$

Proof. We prove that Inequality (15) holds. Based on Lemma 3, we obtain Equation (16), shown at the bottom of the page. Consider the following two cases.

1) When $0 < P(1) < \frac{1}{2}$, we further obtain

$$\begin{aligned}
& (1 - P(1)) \log_2 \left[\frac{3}{4} (1 - P(1)) \right] + P(2) \log_2 \frac{8}{3} \\
&\leq (1 - P(1)) \log_2 \left[\frac{3}{4} (1 - P(1)) \right] + P(1) \log_2 \frac{8}{3} \\
&= (1 - P(1)) \log_2 (1 - P(1)) + \log_2 \frac{3}{4} + P(1) \log_2 \frac{32}{9} \\
&\stackrel{(a)}{<} \left(1 - \frac{1}{2} \right) \log_2 \left(1 - \frac{1}{2} \right) + \log_2 \frac{3}{4} + \frac{1}{2} \log_2 \frac{32}{9} \\
&= 0,
\end{aligned}$$

where (a) is true because $(1-x) \log_2 (1-x) + \log_2 \frac{3}{4} + x \log_2 \frac{32}{9}$ is strictly monotonically increasing over the interval $(0, \frac{1}{2})$.

2) When $\frac{1}{2} \leq P(1) < 1$, we further obtain

$$\begin{aligned} & (1 - P(1)) \log_2 \left[\frac{3}{4}(1 - P(1)) \right] + P(2) \log_2 \frac{8}{3} \\ & \leq P(2) \log_2 \left[\frac{3}{4}(1 - P(1)) \right] + P(2) \log_2 \frac{8}{3} \\ & = P(2) \log_2 [2(1 - P(1))] \\ & \leq P(2) \log_2 \left[2 \left(1 - \frac{1}{2} \right) \right] \\ & = 0. \end{aligned}$$

□

C. Our research approach concerning the upper bound of C^*

In this subsection, we present our approach for studying the upper bound of C^* based on the tighter probability inequality proposed in Lemma 2.

Inequality (11) can equivalently be rewritten as follows:

$$\begin{aligned} & \sum_{a=3}^{\infty} P(a) [\log_2 a - \log_2 (1 - P(1))] \\ & \leq H(P) + P(1) \log_2 P(1) + P(2) \log_2 (1 - P(1)) \\ & \quad + \sum_{a=2}^{\infty} P(a+1) \log_2 \left(1 + \frac{1}{a} \right) \left(1 - \frac{1}{a} \right)^{a-1}. \end{aligned} \quad (17)$$

On the basis of the research approaches summarized in Subsection IV-A, we improve upon the three key points involved in the proof of the upper bound of C^* according to Inequality (17). First, the codeword length upper bound inequality is improved from the traditional $L_C(a) \leq K_1 + K_2 \log_2 a$ to

$$L_C(a) \leq F_2(P(1)) [\log_2 a - \log_2 (1 - P(1))], \quad (18)$$

where $F_2(P(1))$ represents a piecewise function. This idea is inspired by the form on the left-hand side of Inequality (17). At this time, Inequality (18) can be more tightly scaled according to different values of $P(1)$. Second, we use Inequality (17), which is tighter than Inequality (9), as the probability inequality for decreasing distributions. Since an infinite summation term is located on the right-hand side of Inequality (17), which

is inconvenient for the analysis, an appropriate scaling process will also be performed at the end of this subsection. Finally, owing to the form on the right-hand side of Inequality (17), we should ultimately obtain

$$\frac{A_P(L_C)}{\max\{1, H(P)\}} \leq F_3(H(P), P(1), P(2))$$

where $F_3(H(P), P(1), P(2))$ is a function of $H(P)$, $P(1)$ and $P(2)$.⁶ This upper bound function F_3 is related not only to $P(1)$ but also to $P(2)$, making the analysis more precise.

The codeword length upper bound inequality is given after the code construction procedure. At the end of this subsection, we provide a probability inequality for decreasing distributions that is convenient for analysis purposes; its form is roughly

$$\begin{aligned} & \sum_{a=3}^{\infty} P(a) [\log_2 a - \log_2 (1 - P(1))] \\ & \leq H(P) + F_4(P(1)) + P(1)F_5(P(1)) + P(2)F_6(P(1)). \end{aligned}$$

The specific form is given by the following lemma.

Lemma 5. *The probability inequality*

$$\begin{aligned} & \sum_{a=3}^{\infty} P(a) [\log_2 a - \log_2 (1 - P(1))] \\ & \leq H(P) + \log_2 C_L + P(1) (\log_2 P(1) - \log_2 C_L) \\ & \quad + P(2) \left[\log_2 (1 - P(1)) + \sum_{a=2}^{L-1} \log_2 C_a - (L-1) \log_2 C_L \right] \end{aligned}$$

holds for any $P \in \mathcal{P}_A$ and all $2 \leq L \in \mathbb{N}$. In particular, when $L = 2$,

$$\begin{aligned} & \sum_{a=3}^{\infty} P(a) [\log_2 a - \log_2 (1 - P(1))] \\ & \leq H(P) + \log_2 \frac{3}{4} + P(1) \left(\log_2 P(1) - \log_2 \frac{3}{4} \right) \\ & \quad + P(2) \left[\log_2 (1 - P(1)) - \log_2 \frac{3}{4} \right] \end{aligned}$$

for any $P \in \mathcal{P}_A$.

⁶In the subsequent proof described in this paper, using Equation (7), $F_3(H(P), P(1), P(2))$ can be further simplified to $F_3(P(1), P(2))$; that is, F_3 is independent of $H(P)$.

$$\begin{aligned} & (1 - P(1)) \log_2 (1 - P(1)) + P(2) + \sum_{n=2}^{\infty} P(n+1) \log_2 \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} \\ & = P(2) \log_2 [2(1 - P(1))] + \sum_{n=2}^{\infty} P(n+1) \log_2 \left[\left(1 - P(1) \right) \left(1 + \frac{1}{n} \right) \left(1 - \frac{1}{n} \right)^{n-1} \right] \\ & \leq P(2) \log_2 [2(1 - P(1))] + \sum_{n=2}^{\infty} P(n+1) \log_2 \left[\left(1 - P(1) \right) \left(1 + \frac{1}{2} \right) \left(1 - \frac{1}{2} \right) \right] \\ & = P(2) \log_2 [2(1 - P(1))] + (1 - P(1) - P(2)) \log_2 \left[\frac{3}{4}(1 - P(1)) \right] \\ & = (1 - P(1)) \log_2 \left[\frac{3}{4}(1 - P(1)) \right] + P(2) \log_2 \frac{8}{3}. \end{aligned} \quad (16)$$

Proof. Based on Lemma 2 and Lemma 3, we obtain

$$\begin{aligned}
& \sum_{n=2}^{\infty} P(n)A_n \\
& \leq P(2) + \sum_{a=2}^{\infty} P(a+1) \log_2 C_a \\
& \leq P(2) + \sum_{a=2}^{L-1} P(a+1) \log_2 C_a + \sum_{a=L}^{\infty} P(a+1) \log_2 C_L \\
& = P(2) + \sum_{a=2}^{L-1} P(a+1) \log_2 C_a + \left(1 - \sum_{a=1}^L P(a)\right) \log_2 C_L \\
& = (1 - P(1)) \log_2 C_L + P(2)(1 - \log_2 C_L) \\
& \quad + \sum_{a=2}^{L-1} P(a+1)(\log_2 C_a - \log_2 C_L) \\
& \leq (1 - P(1)) \log_2 C_L + P(2)(1 - \log_2 C_L) \\
& \quad + \sum_{a=2}^{L-1} P(2)(\log_2 C_a - \log_2 C_L) \\
& = (1 - P(1)) \log_2 C_L + P(2) \left(1 + \sum_{a=2}^{L-1} \log_2 C_a - (L-1) \log_2 C_L\right).
\end{aligned}$$

Furthermore, we have that

$$\begin{aligned}
& \sum_{a=2}^{\infty} P(a) [\log_2 a - \log_2 (1 - P(1))] \\
& \leq H(P) + P(1) \log_2 P(1) + (1 - P(1)) \log_2 C_L \\
& \quad + P(2) \left(1 + \sum_{a=2}^{L-1} \log_2 C_a - (L-1) \log_2 C_L\right),
\end{aligned}$$

and hence, we can obtain

$$\begin{aligned}
& \sum_{a=3}^{\infty} P(a) [\log_2 a - \log_2 (1 - P(1))] \\
& \leq H(P) + P(1) \log_2 P(1) + (1 - P(1)) \log_2 C_L \\
& \quad + P(2) \left[1 + \sum_{a=2}^{L-1} \log_2 C_a - (L-1) \log_2 C_L - 1 + \log_2 (1 - P(1))\right] \\
& = H(P) + \log_2 C_L + P(1)(\log_2 P(1) - \log_2 C_L) \\
& \quad + P(2) \left[\log_2 (1 - P(1)) + \sum_{a=2}^{L-1} \log_2 C_a - (L-1) \log_2 C_L\right].
\end{aligned}$$

□

V. A TIGHTER UPPER BOUND FOR C^* AND THE CORRESPONDING CODEWORD LENGTH CONSTRUCTION SCHEMES

In this section, we construct a new class of UCIs, which is termed the ν code. The ν code has an expansion factor $C_\nu = 2.0386$. This leads to

$$\begin{aligned}
C^* &= \inf_{C \in \mathcal{UCI}} C_C^* \\
&\leq C_\nu^* \\
&= \inf \left\{ C_\nu \left| \frac{A_P(L_C)}{\max\{1, H(P)\}} \leq C_\nu \text{ for } \forall P \in \mathcal{P}_A \right. \right\} \quad (19) \\
&\leq 2.0386.
\end{aligned}$$

We first present the length function of the ν code and then provide its specific construction process.

A. The length function of the ν code and its existence

The goal of constructing the ν code is to ensure that the length of its codewords can satisfy the codeword length upper bound inequality in the form of Equation (18). Therefore, we directly define the codeword length of the ν code as follows. For all $a \in \mathcal{A}$, the length function $L_\nu(a)$ is defined as

$$L_\nu(a) \triangleq L_\delta(a) + \Delta(a), \quad (20)$$

where

$$\Delta(a) \triangleq \begin{cases} 2, & \text{if } a = 6, 7, \\ 1, & \text{if } a = 3, 4, 5, \\ -1, & \text{if } a = 2 \text{ or } \lfloor \log_2 a \rfloor \in S_{\nu 1}, \\ -2, & \text{if } \lfloor \log_2 a \rfloor \in S_{\nu 2}, \\ 0, & \text{otherwise,} \end{cases}$$

$$S_{\nu 1} \triangleq \{7, 15, 16, \dots, 24, 37, 38, \dots, 50, 68, 69, \dots, 84\},$$

$$S_{\nu 2} \triangleq \{31, 32, 33, 34, 35, 36, 63, 64, 65, 66, 67\}.$$

Next, we prove the existence of a prefix code with such codeword lengths. More precisely, we prove the following theorem.

Theorem 3. *There exist prefix codes whose codeword lengths are $L_\nu(1), L_\nu(2), \dots, L_\nu(a), \dots$*

Proof. From Kraft's inequality [35], [36], this is equivalent to proving that $\sum_{a=1}^{\infty} 2^{-L_\nu(a)} \leq 1$. Our objective is to prove that

$$\sum_{a=1}^{\infty} 2^{-L_\nu(a)} = 1 \quad (21)$$

holds. First, we calculate

$$\begin{aligned}
\sum_{a=1}^{\infty} 2^{-L_\delta(a)} &= \sum_{a=1}^{\infty} 2^{-(1 + \lfloor \log_2 a \rfloor + 2 \lfloor \log_2 (1 + \lfloor \log_2 a \rfloor) \rfloor)} \\
&\stackrel{(b)}{=} \sum_{t=0}^{\infty} 2^t \cdot 2^{-(1+t+2 \lfloor \log_2 (1+t) \rfloor)} \\
&= 2^{-1} \sum_{t=0}^{\infty} \left(\frac{1}{4}\right)^{\lfloor \log_2 (1+t) \rfloor} \\
&\stackrel{(c)}{=} 2^{-1} \sum_{s=0}^{\infty} 2^s \left(\frac{1}{4}\right)^s \\
&= 2^{-1} \sum_{s=0}^{\infty} 2^{-s} \\
&= 1,
\end{aligned} \quad (22)$$

where (b) and (c) are the results of the substitutions $t = \lfloor \log_2 a \rfloor$ and $s = \lfloor \log_2 (1+t) \rfloor$, respectively. Due to Equation (22) and the definition of $L_\nu(a)$, proving Equation (21) is equivalent to proving that

$$\begin{aligned}
& \sum_{a=2}^7 2^{-L_\delta(a)} + \sum_{\lfloor \log_2 a \rfloor \in S_{\nu 1} \cup S_{\nu 2}} 2^{-L_\delta(a)} \\
&= \sum_{a=2}^7 2^{-L_\nu(a)} + \sum_{\lfloor \log_2 a \rfloor \in S_{\nu 1} \cup S_{\nu 2}} 2^{-L_\nu(a)}. \quad (23)
\end{aligned}$$

Afterward, we obtain

$$\begin{aligned}
& \sum_{a=2}^7 2^{-L_\delta(a)} + \sum_{[\log_2 a] \in S_{\nu 1} \cup S_{\nu 2}} 2^{-L_\delta(a)} \\
&= 2 \cdot 2^{-4} + 4 \cdot 2^{-5} + 2^7 \cdot 2^{-14} + \sum_{t=15}^{24} 2^t \cdot 2^{-(1+t+2\lfloor \log_2(1+t) \rfloor)} \\
&+ \sum_{t=31}^{50} 2^t \cdot 2^{-(1+t+2\lfloor \log_2(1+t) \rfloor)} + \sum_{t=63}^{84} 2^t \cdot 2^{-(1+t+2\lfloor \log_2(1+t) \rfloor)} \\
&= 2^{-2} + 2^{-7} + \sum_{t=15}^{24} 2^{-9} + \sum_{t=31}^{50} 2^{-11} + \sum_{t=63}^{84} 2^{-13} \\
&= \frac{1187}{4096},
\end{aligned}$$

and

$$\begin{aligned}
& \sum_{a=2}^7 2^{-L_\nu(a)} + \sum_{[\log_2 a] \in S_{\nu 1} \cup S_{\nu 2}} 2^{-L_\nu(a)} \\
&= 2^{-3} + 2 \cdot 2^{-5} + 2^{-6} + 2^7 \cdot 2^{-13} + \sum_{t=15}^{24} 2^t \cdot 2^{-(t+2\lfloor \log_2(1+t) \rfloor)} \\
&+ \sum_{t=31}^{36} 2^t \cdot 2^{-(1+t+2\lfloor \log_2(1+t) \rfloor)} + \sum_{t=37}^{50} 2^t \cdot 2^{-(t+2\lfloor \log_2(1+t) \rfloor)} \\
&+ \sum_{t=63}^{67} 2^t \cdot 2^{-(1+t+2\lfloor \log_2(1+t) \rfloor)} + \sum_{t=68}^{84} 2^t \cdot 2^{-(t+2\lfloor \log_2(1+t) \rfloor)} \\
&= 2^{-3} + 3 \cdot 2^{-5} + \sum_{t=15}^{24} 2^{-8} + \sum_{t=31}^{36} 2^{-9} + \sum_{t=37}^{50} 2^{-10} + \sum_{t=63}^{67} 2^{-11} + \sum_{t=68}^{84} 2^{-12} \\
&= \frac{1187}{4096}.
\end{aligned}$$

Thus, Equation (23) holds. The proof is complete. $\square$

Finally, the main theorem of this paper is presented here, with its proof provided in the next section.

Theorem 4. *The ν code is a UCI and has an expansion factor $C_\nu = 2.0386$; that is,*

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq 2.0386$$

for all $P \in \mathcal{P}_A$. Thus, $C^* \leq 2.0386$.

B. The specific construction of the ν code

In this subsection, we construct the ν code using two methods.

Because the length function of the ν code is similar to that of the Elias δ code, the first construction scheme is similar to the way in which Elias constructed the δ code. Let us first construct an auxiliary $\Delta\gamma$ code. Since the newly constructed code is obtained from a change in the Elias γ code, it is called the $\Delta\gamma$ code. Notably, the $\Delta\gamma$ code is encoded starting from 0. The encoding strategy of the $\Delta\gamma$ code is as follows. The first 86 codewords of the $\Delta\gamma$ code are shown in Table IV, and from the 87-th codeword of the encoding onward, $\Delta\gamma(a) = \gamma(a)$ for all $a \geq 86$. The last column of Table IV shows the relationship between the γ code and the $\Delta\gamma$ code, where

$\gamma(32) \cap \gamma(33)$ denotes the shortening of $\gamma(32) = 00000100000$ and $\gamma(33) = 00000100001$ refers to the common longest prefix 0000010000.

ν code: $\mathcal{A} \rightarrow \{0, 1\}^*$ can be represented as

$$\nu(a) \triangleq \begin{cases} 1, & \text{if } a = 1, \\ 010, & \text{if } a = 2, \\ 01100, & \text{if } a = 3, \\ 011010, & \text{if } a = 4, \\ 011011, & \text{if } a = 5, \\ 0111000, & \text{if } a = 6, \\ 0111001, & \text{if } a = 7, \\ \Delta\gamma(|\beta(a)|)[\beta(a)], & \text{if } a \geq 8. \end{cases} \quad (24)$$

From the construction of the ν code, it can be seen that the auxiliary $\Delta\gamma$ code is only activated when $a \geq 8$, which indicates that the $\Delta\gamma$ code starts affecting the construction process of the ν code from the $5 = 1 + (\log_2 8 + 1)$ -th codeword 01111. It can be verified that the constructed ν code satisfies Equation (20). From the construction schemes of the $\Delta\gamma$ code and ν code, Equation (20) is clearly satisfied when $a \leq 7$ or $a \geq 2^{85}$. Therefore, it is only necessary to verify these 82 values when $a = 2^t$, where $t \in \{3, 4, \dots, 84\}$. For example, when $a = 2^{33}$, we obtain

$$L_\nu(2^{33}) = 1 + 33 + 2\lfloor \log_2(1 + 33) \rfloor - 2 = 42$$

due to Equation (20), and $\nu(2^{33}) = 000011000\underbrace{00 \dots 0}_{33}$ based

on the construction of the ν code. This verification is not obvious. Therefore, we provide a second construction process.

The second construction process originates from alphabetic codes [38]–[40]. Nakatsu [38] proposed a scheme for constructing alphabetic codes, which is described as follows.⁷ Let l_n denote the length of the n -th codeword. Let $\alpha_n \triangleq \min\{l_{n-1}, l_n\}$ for all $2 \leq n \in \mathbb{N}$. We define $\text{trunc}(x, n) \triangleq \frac{\lfloor 2^n x \rfloor}{2^n}$, which denotes the first n bits after the decimal point of the binary representation of the real number $x < 1$. The coding scheme takes the first l_n bits after the decimal point of the binary representation of $\text{sum}(n)$ as the n -th codeword for all $n \in \mathbb{N}$, where

$$\text{sum}(n) \triangleq \begin{cases} 0, & \text{if } n = 1, \\ \text{trunc}(\text{sum}(n-1), \alpha_n) + \left(\frac{1}{2}\right)^{\alpha_n}, & \text{if } n \geq 2. \end{cases}$$

In this paper, the codeword length satisfies $l_{n-1} \leq l_n$ for all $2 \leq n \in \mathbb{N}$; therefore, we can obtain that $\alpha_n = l_{n-1}$ for all $2 \leq n \in \mathbb{N}$ and

$$\text{sum}(n) = \begin{cases} 0, & \text{if } n = 1, \\ \sum_{i=1}^{n-1} \left(\frac{1}{2}\right)^{l_i}, & \text{if } n \geq 2. \end{cases}$$

Taking the ν code as an example, Equation (20) needs to be satisfied. Therefore, we have that $l_1 = 1, l_2 = 3, l_3 = 5, l_4 = 6, l_5 = 6, l_6 = 7, l_7 = 8, l_8 = 8$. By the definition of $\text{sum}(n)$, we obtain $\text{sum}(1) = 0$, $\text{sum}(2) = \frac{1}{2} = (0.1)_2$, $\text{sum}(3) = \frac{5}{8} = (0.101)_2$, $\text{sum}(4) = \frac{21}{32} = (0.10101)_2$, $\text{sum}(5) = \frac{43}{64} = (0.101011)_2$, $\text{sum}(6) = \frac{11}{16} = (0.1011)_2$, $\text{sum}(7) = \frac{89}{128} =$

⁷Although the scheme developed in [38] targets a finite alphabet, any two codewords constructed by the scheme are not prefixes of each other; thus, extending this alphabet to a countably infinite alphabet is feasible.

TABLE IV
THE FIRST 85 CODEWORDS OF THE γ CODE AND THE FIRST 86 CODEWORDS OF THE $\Delta\gamma$ CODE

n	γ code	$\Delta\gamma$ code	relationship	n	γ code	$\Delta\gamma$ code	relationship
0	—	1	$\gamma(1)$	43	000001 01011	000011110 1	$\gamma(30)1$
1	1	010	$\gamma(2)$	44	000001 01100	000011111 0	$\gamma(31)0$
2	01 0	011 0	$\gamma(3)0$	45	000001 01101	000011111 1	$\gamma(31)1$
3	01 1	011 100	$\gamma(3)100$	46	000001 01110	0000010000	$\gamma(32) \cap \gamma(33)$
4	001 00	011 11	$\gamma(3)11$	47	000001 01111	0000010001	$\gamma(34) \cap \gamma(35)$
5	001 01	00100	$\gamma(4)$	48	000001 10000	0000010010	$\gamma(36) \cap \gamma(37)$
6	001 10	00101	$\gamma(5)$	49	000001 10001	0000010011	$\gamma(38) \cap \gamma(39)$
7	001 11	00110	$\gamma(6)$	50	000001 10010	0000010100	$\gamma(40) \cap \gamma(41)$
8	0001 000	011 101	$\gamma(3)101$	51	000001 10011	0000010101	$\gamma(42) \cap \gamma(43)$
9	0001 001	00111 00	$\gamma(7)00$	52	000001 10100	00000101100	$\gamma(44)$
10	0001 010	00111 01	$\gamma(7)01$	53	000001 10101	00000101101	$\gamma(45)$
11	0001 011	00111 10	$\gamma(7)10$	54	000001 10110	00000101110	$\gamma(46)$
12	0001 100	00111 11	$\gamma(7)11$	55	000001 10111	00000101111	$\gamma(47)$
13	0001 101	0001000	$\gamma(8)$	56	000001 11000	00000110000	$\gamma(48)$
14	0001 110	0001001	$\gamma(9)$	57	000001 11001	00000110001	$\gamma(49)$
15	0001 111	0001010	$\gamma(10)$	58	000001 11010	00000110010	$\gamma(50)$
16	00001 0000	0001011 0	$\gamma(11)0$	59	000001 11011	00000110011	$\gamma(51)$
17	00001 0001	0001011 1	$\gamma(11)1$	60	000001 11100	00000110100	$\gamma(52)$
18	00001 0010	0001100 0	$\gamma(12)0$	61	000001 11101	00000110101	$\gamma(53)$
19	00001 0011	0001100 1	$\gamma(12)1$	62	000001 11110	00000110110	$\gamma(54)$
20	00001 0100	0001101 0	$\gamma(13)0$	63	000001 11111	00000110111	$\gamma(55)$
21	00001 0101	0001101 1	$\gamma(13)1$	64	0000001 000000	00000111000	$\gamma(56)$
22	00001 0110	0001110 0	$\gamma(14)0$	65	0000001 000001	00000111001	$\gamma(57)$
23	00001 0111	0001110 1	$\gamma(14)1$	66	0000001 000010	00000111010	$\gamma(58)$
24	00001 1000	0001111 0	$\gamma(15)0$	67	0000001 000011	00000111011	$\gamma(59)$
25	00001 1001	0001111 1	$\gamma(15)1$	68	0000001 000100	00000111100	$\gamma(60)$
26	00001 1010	000010000	$\gamma(16)$	69	0000001 000101	00000111101 0	$\gamma(61)0$
27	00001 1011	000010001	$\gamma(17)$	70	0000001 000110	00000111101 1	$\gamma(61)1$
28	00001 1100	000010010	$\gamma(18)$	71	0000001 000111	00000111110 0	$\gamma(62)0$
29	00001 1101	000010011	$\gamma(19)$	72	0000001 001000	00000111110 1	$\gamma(62)1$
30	00001 1110	000010100	$\gamma(20)$	73	0000001 001001	00000111111 0	$\gamma(63)0$
31	00001 1111	000010101	$\gamma(21)$	74	0000001 001010	00000111111 1	$\gamma(63)1$
32	000001 00000	000010110	$\gamma(22)$	75	0000001 001011	000000100000	$\gamma(64) \cap \gamma(65)$
33	000001 00001	000010111	$\gamma(23)$	76	0000001 001100	000000100001	$\gamma(66) \cap \gamma(67)$
34	000001 00010	000011000	$\gamma(24)$	77	0000001 001101	000000100010	$\gamma(68) \cap \gamma(69)$
35	000001 00011	000011001	$\gamma(25)$	78	0000001 001110	000000100011	$\gamma(70) \cap \gamma(71)$
36	000001 00100	000011010	$\gamma(26)$	79	0000001 001111	000000100100	$\gamma(72) \cap \gamma(73)$
37	000001 00101	000011011	$\gamma(27)$	80	0000001 010000	000000100101	$\gamma(74) \cap \gamma(75)$
38	000001 00110	000011100 0	$\gamma(28)0$	81	0000001 010001	000000100110	$\gamma(76) \cap \gamma(77)$
39	000001 00111	000011100 1	$\gamma(28)1$	82	0000001 010010	000000100111	$\gamma(78) \cap \gamma(79)$
40	000001 01000	000011101 0	$\gamma(29)0$	83	0000001 010011	000000101000	$\gamma(80) \cap \gamma(81)$
41	000001 01001	000011101 1	$\gamma(29)1$	84	0000001 010100	000000101001	$\gamma(82) \cap \gamma(83)$
42	000001 01010	000011110 0	$\gamma(30)0$	85	0000001 010101	000000101010	$\gamma(84) \cap \gamma(85)$

$(0.1011001)_2$ and $sum(8) = \frac{45}{64} = (0.101101)_2$. We take the first l_n bits after the decimal point of the binary representation of $sum(n)$ as the n -th codeword; thus, we obtain $\nu(1) = 0$, $\nu(2) = 100$, $\nu(3) = 10100$, $\nu(4) = 101010$, $\nu(5) = 101011$, $\nu(6) = 1011000$, $\nu(7) = 1011001$ and $\nu(8) = 10110100$.

From the first 8 codewords, it can be seen that the codes obtained by the two construction schemes are not the same, but both of them satisfy Equation (20). The advantage of the first construction scheme is its high coding efficiency; for a large integer n , $\nu(n)$ can be obtained relatively quickly. The advantage of the second construction scheme is that it can be easily verified to satisfy Equation (20).

VI. THE COMPLETE PROOF OF THEOREM 4

The complete proof of Theorem 4 is given in this section. We divide the proof into the following two subsections. We first need to prove the codeword length upper bound inequalities for the ν code.

A. The codeword length upper bound inequalities for the ν code

We provide the codeword length upper bound inequalities for the ν code, as shown in the following lemma. Notably, in our research approach, to carry out a more precise inequality scaling process for the codeword length, F_2 in Inequality (18) is a piecewise function. Therefore, the following lemma is divided into several cases based on the value of $P(1)$.

Lemma 6. Suppose that $P \in \mathcal{P}_A$ is any given probability distribution. Let

$$g_{(c_1, c_2)}(a, P(1)) \triangleq \frac{c_1 \cdot (\log_2 a - \log_2 (1 - P(1)))}{c_2 - \log_2 (1 - P(1))}. \quad (25)$$

- 1) When $0 < P(1) \leq 1 - 3^6 \cdot 2^{-10} \approx 0.28809$, $L_\nu(a) \leq g_{(5, \log_2 3)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;
- 2) when $1 - 3^6 \cdot 2^{-10} < P(1) \leq 0.5$, $L_\nu(a) \leq g_{(6, 2)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;
- 3) when $0.5 < P(1) \leq 1 - 2^{-\frac{19}{7}} \approx 0.84762$, $L_\nu(a) \leq g_{(8, 3)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;

- 4) when $1 - 2^{-\frac{19}{7}} < P(1) \leq 1 - 2^{-\frac{41}{8}} \approx 0.97134$, $L_\nu(a) \leq g_{(15,8)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;
- 5) when $1 - 2^{-\frac{41}{8}} < P(1) \leq 1 - 2^{-\frac{65}{11}} \approx 0.98336$, $L_\nu(a) \leq g_{(23,15)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;
- 6) when $1 - 2^{-\frac{65}{11}} < P(1) \leq 1 - 2^{-\frac{83}{13}} \approx 0.98803$, $L_\nu(a) \leq g_{(34,25)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;
- 7) when $1 - 2^{-\frac{83}{13}} < P(1) \leq 1 - 2^{-\frac{103}{15}} \approx 0.99143$, $L_\nu(a) \leq g_{(47,37)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;
- 8) when $1 - 2^{-\frac{103}{15}} < P(1) \leq 1 - 2^{-\frac{68}{9}} \approx 0.99468$, $L_\nu(a) \leq g_{(62,51)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$;
- 9) when $1 - 2^{-\frac{68}{9}} < P(1) \leq 1 - 2^{-\frac{94}{11}} \approx 0.99732$, $L_\nu(a) \leq g_{(98,85)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$; and
- 10) when $1 - 2^{-\frac{94}{11}} < P(1) \leq 1 - 2^{-\frac{833}{65}} \approx 0.99986$, $L_\nu(a) \leq g_{(142,127)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$.
- 11) The length function

$$L_\nu(a) \leq \frac{65}{64} \left(\frac{833}{65} + \log_2 a \right) = \frac{833}{64} + \frac{65}{64} \log_2 a$$

for all $3 \leq a \in \mathcal{A}$.

Proof. We first outline the proof strategy. According to the construction of Equation (24), we proceed by considering two cases. When $3 \leq a \leq 7$, we directly examine the monotonicity of $\frac{c_1}{c_2 - \log_2(1 - P(1))} \left(\log_2 a - \log_2(1 - P(1)) \right)$ with respect to $P(1)$, thereby comparing the magnitudes of $\frac{c_1}{c_2 - \log_2(1 - P(1))} \left(\log_2 a - \log_2(1 - P(1)) \right)$ and $L_\nu(a)$ within the corresponding interval. When $8 \leq a \in \mathcal{A}$, $L_\nu(a) \leq g_{(c_1, c_2)}(a, P(1))$ is equivalent to

$$1 + \lfloor \log_2 a \rfloor + 2 \lfloor \log_2(1 + \lfloor \log_2 a \rfloor) \rfloor + \Delta(a) \leq \frac{c_1}{c_2 - \log_2(1 - P(1))} \left(\log_2 a - \log_2(1 - P(1)) \right). \quad (27)$$

Let $t \triangleq \lfloor \log_2 a \rfloor$ and $x \triangleq -\log_2(1 - P(1))$. To prove Equation (27), we only need to show that

$$1 + t + 2 \lfloor \log_2(1 + t) \rfloor + \tilde{\Delta}(t) \leq \frac{c_1}{c_2 + x} (t + x) \iff \left(\frac{c_1}{c_2 + x} - 1 \right) t + \frac{c_1 x}{c_2 + x} - 1 - \tilde{\Delta}(t) \geq 2 \lfloor \log_2(1 + t) \rfloor,$$

where $\tilde{\Delta}(\lfloor \log_2 a \rfloor) \triangleq \Delta(a)^8$ for all $8 \leq a \in \mathcal{A}$. Based on the definition of $\Delta(\cdot)$, $\tilde{\Delta}(t) = 0, -1, -2$ for all $3 \leq t \in \mathcal{A}$. Let

$$h_{(c_1, c_2)}(t, x) \triangleq \left(\frac{c_1}{c_2 + x} - 1 \right) t + \frac{c_1 x}{c_2 + x} - 1. \quad (28)$$

Thus, to show that $L_\nu(a) \leq g_{(c_1, c_2)}(a, P(1))$ for all $8 \leq a \in \mathcal{A}$ and $d_1 < P(1) \leq d_2$, it is only necessary to show that

$$h_{(c_1, c_2)}(t, x) - \tilde{\Delta}(t) \geq 2 \lfloor \log_2(1 + t) \rfloor \quad (29)$$

for all $3 \leq t \in \mathbb{N}$ and $-\log_2(1 - d_1) < x \leq -\log_2(1 - d_2)$.

- 1) To prove that $L_\nu(a) \leq g_{(5, \log_2 3)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$, consider the following four cases.

⁸According to the definition of $\Delta(\cdot)$, if $8 \leq a_1 \neq a_2$ and $\lfloor \log_2 a_1 \rfloor = \lfloor \log_2 a_2 \rfloor$, then $\Delta(a_1) = \Delta(a_2)$ still holds. Therefore, the definition of $\Delta(\cdot)$ is well-defined, and it can also be directly defined as $\tilde{\Delta}(t) \triangleq \Delta(2^t)$.

- a) When $a = 3$, we obtain

$$g_{(5, \log_2 3)}(3, P(1)) = \frac{5 \{ \log_2 3 - \log_2 [1 - P(1)] \}}{\log_2 3 - \log_2 [1 - P(1)]} = L_\nu(3).$$

- b) When $a = 4, 5$, since $L_\nu(4) = L_\nu(5) = 6$ and $g_{(5, \log_2 3)}(a, P(1))$ is a strictly increasing sequence with respect to a , it is only necessary to show that $g_{(5, \log_2 3)}(4, P(1)) \geq L_\nu(4) = 6$. Owing to

$$g_{(5, \log_2 3)}(4, P(1)) = 5 + \frac{10 - 5 \log_2 3}{\log_2 3 - \log_2(1 - P(1))},$$

$g_{(5, \log_2 3)}(4, P(1))$ is strictly monotonically decreasing on the interval $(0, 1 - 3^6 \cdot 2^{-10}]$ with respect to $P(1)$. Thus,

$$g_{(5, \log_2 3)}(4, P(1)) \geq g_{(5, \log_2 3)}(4, 1 - 3^6 \cdot 2^{-10}) = 6 = L_\nu(4).$$

- c) When $a = 6, 7$, since $L_\nu(6) = L_\nu(7) = 7$, similarly, we only need to show that $g_{(5, \log_2 3)}(6, P(1)) \geq L_\nu(6) = 7$. Due to

$$g_{(5, \log_2 3)}(6, P(1)) = 5 + \frac{5}{\log_2 3 - \log_2(1 - P(1))},$$

$g_{(5, \log_2 3)}(6, P(1))$ is strictly monotonically decreasing on the interval $(0, 1 - 3^6 \cdot 2^{-10}]$ with respect to $P(1)$. Thus,

$$g_{(5, \log_2 3)}(6, P(1)) \geq g_{(5, \log_2 3)}(6, 1 - 3^6 \cdot 2^{-10}) > 7 = L_\nu(6).$$

- d) When $a \geq 8$, from Equation (29), we only need to show that

$$h_{(5, \log_2 3)}(t, x) - \tilde{\Delta}(t) \geq 2 \lfloor \log_2(1 + t) \rfloor \quad (30)$$

for all $3 \leq t \in \mathbb{N}$ and $0 < x \leq 10 - 6 \log_2 3$. When $t = 3$, the left-hand side of Equation (30) is

$$h_{(5, \log_2 3)}(3, x) = 1 + \frac{15 - 5 \log_2 3}{x + \log_2 3} \geq 1 + \frac{15 - 5 \log_2 3}{10 - 6 \log_2 3 + \log_2 3} > 4 = 2 \lfloor \log_2(1 + 3) \rfloor.$$

Thereafter, each increase of 2 on the right-hand side of Equation (30) increases t by at least $7 - 3 = 4$; thus, the left-hand side of Equation (30) increases by at least

$$\left(\frac{5}{\log_2 3 + x} - 1 \right) \times 4 \geq \left(\frac{5}{\log_2 3 + 10 - 6 \log_2 3} - 1 \right) \times 4 > 4.$$

Therefore, Equation (30) holds when $3 \leq t \in \mathbb{N}$ and $0 < x \leq 10 - 6 \log_2 3$.

- 2)-10) The proof strategy employed for these cases is exactly the same as 1). To avoid repetition and improve the readability of this paper, we move these proofs to Appendix A.

11) When $3 \leq a \leq 255$, we obtain

$$L_\nu(a) \leq L_\nu(255) = 13 < \frac{833}{64} + \frac{65}{64} \log_2 a.$$

When $a \geq 256$, let $t \triangleq \lfloor \log_2 a \rfloor \geq 8$; then, $L_\nu(a) = L_\delta(a) + \Delta(a) \leq L_\delta(a)$. Therefore, we only need to show that

$$\begin{aligned} 1 + t + 2\lfloor \log_2(1+t) \rfloor &\leq \frac{833}{64} + \frac{65}{64}t \\ \iff \frac{t}{64} + \frac{769}{64} &\geq 2\lfloor \log_2(1+t) \rfloor. \end{aligned} \quad (31)$$

We consider the following four cases.

a) When $8 \leq t \leq 14$, the left-hand side of Equation (31) is

$$\frac{t}{64} + \frac{769}{64} \geq \frac{8}{64} + \frac{769}{64} > 12 > 6 = 2\lfloor \log_2(1+t) \rfloor.$$

b) When $15 \leq t \leq 30$, the left-hand side of Equation (31) is

$$\frac{t}{64} + \frac{769}{64} \geq \frac{15}{64} + \frac{769}{64} = 12.25 > 8 = 2\lfloor \log_2(1+t) \rfloor.$$

c) When $31 \leq t \leq 62$, the left-hand side of Equation (31) is

$$\frac{t}{64} + \frac{769}{64} \geq \frac{31}{64} + \frac{769}{64} = 12.5 > 10 = 2\lfloor \log_2(1+t) \rfloor.$$

d) When $63 \leq t \leq 126$, the left-hand side of Equation (31) is

$$\frac{t}{64} + \frac{769}{64} \geq \frac{63}{64} + \frac{769}{64} = 13 > 12 = 2\lfloor \log_2(1+t) \rfloor.$$

e) We prove that Equation (31) holds for all $t \geq 127$. When $t = 127$, the left-hand side of Equation (31) is

$$\frac{t}{64} + \frac{769}{64} = \frac{127}{64} + \frac{769}{64} = 14 = 2\lfloor \log_2(1+t) \rfloor.$$

Thereafter, each increase of 2 on the right-hand side of Equation (31) increases t by at least $255 - 127 = 128$, and thus, the left-hand side of Equation (31) increases by at least $\frac{1}{64} \times 128 = 2$. Therefore, Equation (31) holds when $127 \leq t \in \mathbb{N}$. $\square$

Remark 1. Lemma 6 is divided into 11 cases on the basis of the values of $P(1)$, mainly because of the joint effect of its codeword construction process and Inequality (18) on the research approach. To achieve the extension factor $C_\nu = 2.0386$ of the ν code in Theorem 4, the above cases cannot be merged.

However, suppose that the requirements concerning the results can be relaxed, for example, by retaining the first four cases of Lemma 6 and proving that the inequality

$$L_\nu(a) \leq \frac{8}{7} \left(\frac{41}{8} + \log_2 a \right)$$

holds for all $3 \leq a \in \mathcal{A}$. Then, the codeword length upper bound inequality needs to be divided into only 5 cases, and it can subsequently be proven that the ν code has an expansion factor $C_\nu = \frac{15}{7} \approx 2.1429$.

B. The proof of Theorem 4

In this subsection, we prove the main theorem of this paper, namely, Theorem 4. To better illustrate the proof, we define some notations in Equation (26), shown at the bottom of the page.

The most important theorem of this paper is proven as follows.

Theorem 5. (Theorem 4 Restated) The ν code is a UCI and has an expansion factor $C_\nu = 2.0386$; that is,

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq 2.0386$$

for all $P \in \mathcal{P}_\mathcal{A}$. Thus, $C^* \leq 2.0386$.

Proof. According to Lemma 6 and Lemma 5, the average codeword length of the ν code is

$$\begin{aligned} A_P(L_\nu) &= P(1) + 3P(2) + \sum_{a=3}^{\infty} P(a)L_\nu(a) \\ &\leq P(1) + 3P(2) + D_{(c_1, c_2)}(P(1)) \sum_{a=3}^{\infty} P(a) \left[\log_2 a - \log_2(1-P(1)) \right] \\ &\leq P(1) + 3P(2) \\ &\quad + D_{(c_1, c_2)}(P(1)) \left[H(P) + \log_2 C_L + P(1)(\log_2 P(1) - \log_2 C_L) \right. \\ &\quad \left. + P(2) \left(\log_2(1-P(1)) + \sum_{n=2}^{L-1} \log_2 C_n - (L-1) \log_2 C_L \right) \right] \\ &= D_{(c_1, c_2)}(P(1))H(P) + D_{(c_1, c_2)}(P(1)) \log_2 C_L \\ &\quad + P(1) \left[1 + D_{(c_1, c_2)}(P(1))(\log_2 P(1) - \log_2 C_L) \right] \\ &\quad + P(2) \left[3 + D_{(c_1, c_2)}(P(1))(\log_2(1-P(1)) \right. \\ &\quad \left. + \sum_{n=2}^{L-1} \log_2 C_n - (L-1) \log_2 C_L) \right] \\ &= Q_{(c_1, c_2)}(P(1), P(2), L) + D_{(c_1, c_2)}(P(1))H(P). \end{aligned}$$

$$\begin{aligned} D_{(c_1, c_2)}(P(1)) &\triangleq \frac{c_1}{c_2 - \log_2(1-P(1))}, \\ J_{(c_1, c_2)}(P(1), L) &\triangleq D_{(c_1, c_2)}(P(1)) \log_2 C_L + P(1) \left[1 + D_{(c_1, c_2)}(P(1))(\log_2 P(1) - \log_2 C_L) \right], \\ R_{(c_1, c_2)}(P(1), L) &\triangleq 3 + D_{(c_1, c_2)}(P(1)) \left(\log_2(1-P(1)) + \sum_{n=2}^{L-1} \log_2 C_n - (L-1) \log_2 C_L \right), \\ Q_{(c_1, c_2)}(P(1), P(2), L) &\triangleq J_{(c_1, c_2)}(P(1), L) + P(2)R_{(c_1, c_2)}(P(1), L). \end{aligned} \quad (26)$$

Based on Equation (7), we obtain

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq \frac{Q_{(c_1, c_2)}(P(1), P(2), L) + D_{(c_1, c_2)}(P(1))H(P)}{\max\{1, H(P)\}} \quad (33) \\ & \leq Q_{(c_1, c_2)}(P(1), P(2), L) + D_{(c_1, c_2)}(P(1)). \end{aligned}$$

Next, we analyse the strategy for further bounding the expression $Q_{(c_1, c_2)}(P(1), P(2), L) + D_{(c_1, c_2)}(P(1))$. In expression $Q_{(c_1, c_2)}(P(1), P(2), L) + D_{(c_1, c_2)}(P(1))$, there are only two unknowns: $P(1)$ and $P(2)$. We choose to address $P(2)$. Since

$$\begin{aligned} & Q_{(c_1, c_2)}(P(1), P(2), L) + D_{(c_1, c_2)}(P(1)) \\ & = D_{(c_1, c_2)}(P(1)) + J_{(c_1, c_2)}(P(1), L) + P(2)R_{(c_1, c_2)}(P(1), L), \end{aligned}$$

the treatment of $P(2)$ should account for the positivity or negativity of $R_{(c_1, c_2)}(P(1), L)$. If $R_{(c_1, c_2)}(P(1), L) < 0$, then $P(2)R_{(c_1, c_2)}(P(1), L) < 0$. If $R_{(c_1, c_2)}(P(1), L) \geq 0$, then

$$\begin{aligned} & P(2)R_{(c_1, c_2)}(P(1), L) \\ & \leq \min\{P(1), 1 - P(1)\}R_{(c_1, c_2)}(P(1), L). \end{aligned}$$

In summary, the strategy can be represented as Equation (32), shown at the bottom of the page.

After handling $P(2)$, only $P(1)$ remains as an unknown, and it can be directly addressed through differentiation. On the basis of the numerical value of $P(1)$, we consider the following thirteen cases.

- 1) Case $0 < P(1) \leq 1 - 3^6 \cdot 2^{-10} \approx 0.28809$: From 1) of Lemma 6, we know that $c_1 = 5$, and $c_2 = \log_2 3$. Consider $L = 2$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(5, \log_2 3)}(P(1), P(2), 2) + D_{(5, \log_2 3)}(P(1)) \\ & \stackrel{(a)}{\leq} Q_{(5, \log_2 3)}(P(1), P(1), 2) + D_{(5, \log_2 3)}(P(1)), \end{aligned}$$

where (a) is true because

$$\begin{aligned} & R_{(c_1, c_2)}(P(1), L) \\ & = 3 + D_{(5, \log_2 3)}(P(1)) \left(\log_2(1 - P(1)) - \log_2 \frac{3}{4} \right) > 0 \end{aligned}$$

for all $P(1) \in (0, 1 - 3^6 \cdot 2^{-10}]$. Let $f_1(x) \triangleq Q_{(5, \log_2 3)}(x, x, 2) + D_{(5, \log_2 3)}(x)$. By calculating the derivative, we know that $f_1(x)$ is decreasing and then increasing over the interval $(0, 1 - 3^6 \cdot 2^{-10}]$. Thus, we obtain

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_1(P(1)) \\ & \leq \max\{f_1(0), f_1(1 - 3^6 \cdot 2^{-10})\} \\ & = f_1(0) < 1.8454 \end{aligned}$$

for all $P(1) \in (0, 1 - 3^6 \cdot 2^{-10}]$.

- 2) Case $1 - 3^6 \cdot 2^{-10} < P(1) \leq 0.5$: From 2) of Lemma 6, we know that $c_1 = 6$, and $c_2 = 2$. Consider $L = 2$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq Q_{(6, 2)}(P(1), P(2), 2) + D_{(6, 2)}(P(1)) \\ & \stackrel{(a)}{\leq} Q_{(6, 2)}(P(1), P(1), 2) + D_{(6, 2)}(P(1)), \end{aligned}$$

where (a) is true because

$$\begin{aligned} & R_{(c_1, c_2)}(P(1), L) \\ & = 3 + D_{(6, 2)}(P(1)) \left(\log_2(1 - P(1)) - \log_2 \frac{3}{4} \right) > 0 \end{aligned}$$

for all $P(1) \in (1 - 3^6 \cdot 2^{-10}, 0.5]$. Let $f_2(x) \triangleq Q_{(6, 2)}(x, x, 2) + D_{(6, 2)}(x)$. By calculating the derivative, we know that $f_2(x)$ is strictly increasing over the interval $(1 - 3^6 \cdot 2^{-10}, 0.5]$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_2(P(1)) \leq f_2(0.5) = 2$$

for all $P(1) \in (1 - 3^6 \cdot 2^{-10}, 0.5]$.

- 3) Case $0.5 < P(1) \leq 1 - 2^{-\frac{19}{7}} \approx 0.84762$: From 3) of Lemma 6, we know that $c_1 = 8$, and $c_2 = 3$. If $L = 2$, then

$$\begin{aligned} & R_{(c_1, c_2)}(P(1), L) \\ & = 3 + D_{(8, 3)}(P(1)) \left(\log_2(1 - P(1)) - \log_2 \frac{3}{4} \right). \end{aligned}$$

Let $j_1(x) \triangleq 3 + D_{(8, 3)}(x) (\log_2(1 - x) - \log_2 \frac{3}{4})$. By calculating the derivative, we know that $j_1(x)$ is strictly decreasing over the interval $(0.5, 1 - 2^{-\frac{19}{7}}]$, and the unique zero point over the interval is $x_1 \approx 0.81876$.

- a) When $0.5 < P(1) \leq x_1$, owing to Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(8, 3)}(P(1), P(2), 2) + D_{(8, 3)}(P(1)) \\ & \stackrel{(a)}{\leq} Q_{(8, 3)}(P(1), 1 - P(1), 2) + D_{(8, 3)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(8, 3)}(P(1), 2) = j_1(P(1)) \geq 0$ for all $P(1) \in (0.5, x_1]$. Let $f_3(x) \triangleq Q_{(8, 3)}(x, 1 - x, 2) + D_{(8, 3)}(x)$. By calculating the derivative, we know that $f_3(x)$ is decreasing and then increasing over the interval $(0.5, x_1]$. Thus, we obtain

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_3(P(1)) \\ & \leq \max\{f_3(0.5), f_3(x_1)\} \\ & = f_3(0.5) = 2 \end{aligned}$$

for all $P(1) \in (0.5, x_1]$.

$$Q_{(c_1, c_2)}(P(1), P(2), L) \leq \begin{cases} Q_{(c_1, c_2)}(P(1), 0, L), & \text{if } R_{(c_1, c_2)}(P(1), L) < 0, \\ Q_{(c_1, c_2)}(P(1), \min\{P(1), 1 - P(1)\}, L), & \text{if } R_{(c_1, c_2)}(P(1), L) \geq 0. \end{cases} \quad (32)$$

- b) When $x_1 < P(1) \leq 1 - 2^{-\frac{19}{7}}$, due to Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(8,3)}(P(1), P(2), 2) + D_{(8,3)}(P(1)) \\ & \stackrel{(a)}{\leq} Q_{(8,3)}(P(1), 0, 2) + D_{(8,3)}(P(1)) \\ & = J_{(8,3)}(P(1), 2) + D_{(8,3)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(8,3)}(P(1), 2) = j_1(P(1)) < 0$ for all $P(1) \in (x_1, 1 - 2^{-\frac{19}{7}}]$. Let $f_4(x) \triangleq J_{(8,3)}(x, 2) + D_{(8,3)}(x)$. By calculating the derivative, we know that $f_4(x)$ is strictly increasing over the interval $(x_1, 1 - 2^{-\frac{19}{7}}]$. Thus, we obtain

$$\begin{aligned} \frac{A_P(L_\nu)}{\max\{1, H(P)\}} & \leq f_4(P(1)) \\ & \leq f_4(1 - 2^{-\frac{19}{7}}) < 1.8761 \end{aligned}$$

for all $P(1) \in (x_1, 1 - 2^{-\frac{19}{7}}]$.

- 4)-12) The proof strategy for these cases is exactly the same as that employed for 1)-3). To avoid repetition and improve the readability of this paper, we move these proofs to Appendix B.
- 13) Case $1 - 2^{-\frac{833}{65}} < P(1) < 1$: Because $1 - 2^{-\frac{833}{65}} < P(1) < 1$, we obtain $\log_2 \frac{1}{1-P(1)} > \frac{833}{65}$. From 11) of Lemma 6 and Lemma 5, the average codeword length of the ν code is

$$\begin{aligned} A_P(L_\nu) & = P(1) + 3P(2) + \sum_{i=3}^{\infty} P(i)L_\nu(i) \\ & \leq P(1) + 3P(2) + \frac{65}{64} \sum_{i=3}^{\infty} P(i) \left(\log_2 i + \frac{833}{65} \right) \\ & < P(1) + 3P(2) + \frac{65}{64} \sum_{i=3}^{\infty} P(i) \left[\log_2 i - \log_2 (1 - P(1)) \right] \\ & \leq P(1) + 3P(2) + \frac{65}{64} \left[H(P) + \log_2 \frac{3}{4} \right. \\ & \quad \left. + P(1) \left(\log_2 P(1) - \log_2 \frac{3}{4} \right) \right. \\ & \quad \left. + P(2) \left(\log_2 (1 - P(1)) - \log_2 \frac{3}{4} \right) \right] \\ & \stackrel{(a)}{<} \frac{65}{64} H(P) + \frac{65}{64} \log_2 \frac{3}{4} \\ & \quad + P(1) \left(1 + \frac{65}{64} \log_2 P(1) - \frac{65}{64} \log_2 \frac{3}{4} \right), \end{aligned}$$

where (a) is true because

$$3 + \frac{65}{64} \left(\log_2 (1 - P(1)) - \log_2 \frac{3}{4} \right) < 0$$

for all $P(1) \in (1 - 2^{-\frac{833}{65}}, 1)$. Furthermore, we have that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq \frac{65}{64} \left(1 + \log_2 \frac{3}{4} \right) + P(1) \left(1 + \frac{65}{64} \log_2 P(1) - \frac{65}{64} \log_2 \frac{3}{4} \right) \end{aligned}$$

due to Equation (7). Let $f_{17}(x) \triangleq \frac{65}{64} \left(1 + \log_2 \frac{3}{4} \right) + x \left(1 + \frac{65}{64} \log_2 x - \frac{65}{64} \log_2 \frac{3}{4} \right)$. By calculating the derivative, we know that $f_{17}(x)$ is strictly increasing over the interval $(1 - 2^{-\frac{833}{65}}, 1)$. Thus, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{17}(P(1)) < f_{17}(1) = 2.015625$$

for all $P(1) \in (1 - 2^{-\frac{833}{65}}, 1)$.

In summary, the ν code is a UCI and has an expansion factor $C_\nu = 2.0386$. Therefore, $C^* \leq 2.0386$. $\square$

Remark 2. We address two points regarding the proof of Theorem 5 being divided into 13 cases. First, this is because the codeword length upper bound inequalities given in Lemma 6 are divided into 11 cases. If the codeword length upper bound inequalities in Remark 1, which are divided into only 5 cases, were used, then the proof of Theorem 5 would only need to be divided into 6 cases. However, the results would only show that the ν code has an expansion factor $C_\nu = \frac{15}{7} \approx 2.1429$. Second, it is important to emphasize the novelty of the obtained result. As shown in Table III, even if previously proposed codes were analysed using the proof strategy proposed in this paper, it would still be impossible to obtain an expansion factor smaller than 2.1.

VII. THE LOWER BOUND OF C_ν^* AND COMPARISONS

In this section, the lower bound of C_ν^* is provided, and the minimum expansion factor range of the ν code is compared with those listed in Table III.

First, we explain that $C_\nu^* > 2.023936$. Consider the probability distribution

$$\bar{P}_2(a) = \begin{cases} 0.992886244, & \text{if } a = 1, \\ \frac{1-0.992886244}{2^{132}}, & \text{if } a = 2, 3, \dots, 2^{132} + 1, \\ 0, & \text{otherwise,} \end{cases}$$

for estimating C_ν^* . We obtain

$$\begin{aligned} & \frac{A_{\bar{P}_2}(L_\nu)}{\max\{1, H(\bar{P}_2)\}} \\ & = \frac{0.992886244 + \frac{1-0.992886244}{2^{132}} \times \sum_{a=2}^{2^{132}+1} L_\nu(a)}{H(\bar{P}_2)} \\ & > \frac{0.992886244 + \frac{1-0.992886244}{2^{132}} \cdot 7.891148088 \times 10^{41}}{H(\bar{P}_2)} \\ & > 2.023936. \end{aligned}$$

Therefore, we obtain $C_\nu^* > 2.023936$.

Second, from Theorem 4, we have that $2.023936 < C_\nu^* \leq 2.0386$. As shown in Table III, the best previous result obtained regarding the upper bound of the minimum expansion factor was $C_\nu^* = 2.5$, and the best previous result obtained regarding the lower bound of the minimum expansion factor was $C_\omega^* > 2.1$. Because $C_\nu^* \leq 2.0386 < 2.1$, the ν code constructed herein is currently optimal in terms of the minimum expansion factor.

VIII. CONCLUSIONS

In this paper, we propose a new proof showing that the minimum expansion factor $C_C^* \geq 2$ for any UCI $\mathcal{C}$. We prove a tighter probability inequality for decreasing distributions, which serves as a new tool for studying the properties of UCIs. On the basis of this inequality, we improve the research approach for determining the upper bound of C^* . A new class of UCIs, termed the ν code, is proposed. We prove that $2.023936 < C_\nu^* \leq 2.0386$. Therefore, the ν code is currently optimal in terms of the minimum expansion factor. Since the ν code has an expansion factor $C_\nu = 2.0386$, the upper bound of C^* is reduced from 2.5 to 2.0386. However, the proof of the upper bound of C^* is not elegant because it needs to be divided into 13 cases. Therefore, there are two unresolved issues, as stated below.

- 1) Given a specific structure or the codeword length of a class of UCIs $\mathcal{C}$, is there an algorithm that can quickly provide the exact value of the minimum expansion factor C_C^* or a smaller range?
- 2) Although the range of C^* has been narrowed to $2 \leq C^* \leq 2.0386$, the exact value of C^* is still unknown.

APPENDIX A

PROOF OF THE REMAINING PART OF LEMMA 6

Proof. 2) To prove that $L_\nu(a) \leq g_{(6,2)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$, consider the following four cases.

a) When $a = 3$, we obtain

$$\begin{aligned} g_{(6,2)}(3, P(1)) &= \frac{6\{\log_2 3 - \log_2[1 - P(1)]\}}{2 - \log_2[1 - P(1)]} \\ &= 6 + \frac{6\log_2 3 - 12}{2 - \log_2[1 - P(1)]} \\ &> 6 + \frac{6\log_2 3 - 12}{2 - \log_2(3^6 \times 2^{-10})} \\ &= 5 = L_\nu(3), \end{aligned}$$

for all $P(1) \in (1 - 3^6 \cdot 2^{-10}, 0.5]$.

b) When $a = 4, 5$, since $L_\nu(4) = L_\nu(5) = 6$ and $g_{(6,2)}(a, P(1))$ is a strictly increasing sequence with respect to a , it is only necessary to show that $g_{(6,2)}(4, P(1)) \geq L_\nu(4) = 6$. We have that

$$g_{(6,2)}(4, P(1)) = \frac{6\{2 - \log_2[1 - P(1)]\}}{2 - \log_2[1 - P(1)]} = L_\nu(4).$$

c) When $a = 6, 7$, since $L_\nu(6) = L_\nu(7) = 7$, similarly, we only need to show that $g_{(6,2)}(6, P(1)) \geq L_\nu(6) = 7$. We obtain

$$\begin{aligned} g_{(6,2)}(6, P(1)) &= 6 + \frac{6\log_2 6 - 12}{2 - \log_2[1 - P(1)]} \\ &\geq 6 + \frac{6\log_2 6 - 12}{2 - \log_2(1 - 0.5)} \\ &= 2 + 2\log_2 6 \\ &> 7 = L_\nu(6), \end{aligned}$$

for all $P(1) \in (1 - 3^6 \cdot 2^{-10}, 0.5]$.

d) When $a \geq 8$, according to Equation (29), we only need to show that

$$h_{(6,2)}(t, x) - \tilde{\Delta}(t) \geq 2\lfloor \log_2(1 + t) \rfloor \quad (34)$$

for all $3 \leq t \in \mathbb{N}$ and $10 - 6\log_2 3 < x \leq 1$. We prove that Equation (34) holds when $3 \leq t \in \mathbb{N}$ and $10 - 6\log_2 3 < x \leq 1$ by proving that

$$h_{(6,2)}(t, x) \geq 2\lfloor \log_2(1 + t) \rfloor \quad (35)$$

holds when $3 \leq t \in \mathbb{N}$ and $10 - 6\log_2 3 < x \leq 1$. When $t = 3$, the left-hand side of Equation (35) is

$$\begin{aligned} h_{(6,2)}(3, x) &= \left(\frac{6}{2+x} - 1 \right) \times 3 + \frac{6x}{2+x} - 1 \\ &= 2 + \frac{6}{2+x} \\ &\geq 2 + \frac{6}{2+1} \\ &= 4 = 2\lfloor \log_2(1 + 3) \rfloor. \end{aligned}$$

Thereafter, each increase of 2 on the right-hand side of Equation (35) increases t by at least $7 - 3 = 4$, and thus, the left-hand side of Equation (35) increases by at least

$$\left(\frac{6}{2+x} - 1 \right) \times 4 \geq \left(\frac{6}{2+1} - 1 \right) \times 4 = 4.$$

Therefore, Equation (35) holds when $3 \leq t \in \mathbb{N}$ and $10 - 6\log_2 3 < x \leq 1$.

3) To prove that $L_\nu(a) \leq g_{(8,3)}(a, P(1))$ for all $3 \leq a \in \mathcal{A}$, consider the following four cases.

a) When $a = 3$, we obtain

$$\begin{aligned} g_{(8,3)}(3, P(1)) &= 8 + \frac{8\log_2 3 - 24}{3 - \log_2[1 - P(1)]} \\ &> 8 + \frac{8\log_2 3 - 24}{3 - \log_2(1 - 0.5)} \\ &> 5 = L_\nu(3), \end{aligned}$$

for all $P(1) \in (0.5, 1 - 2^{-\frac{19}{7}}]$.

b) When $a = 4, 5$, since $L_\nu(4) = L_\nu(5) = 6$ and $g_{(8,3)}(a, P(1))$ is a strictly increasing sequence with respect to a , it is only necessary to show that $g_{(8,3)}(4, P(1)) \geq L_\nu(4) = 6$. We have that

$$\begin{aligned} g_{(8,3)}(4, P(1)) &= 8 - \frac{8}{3 - \log_2[1 - P(1)]} \\ &> 8 - \frac{8}{3 - \log_2(1 - 0.5)} \\ &= 6 = L_\nu(4), \end{aligned}$$

for all $P(1) \in (0.5, 1 - 2^{-\frac{19}{7}}]$.

c) When $a = 6, 7$, since $L_\nu(6) = L_\nu(7) = 7$, similarly, we only need to show that $g_{(8,3)}(6, P(1)) \geq L_\nu(6) = 7$. We obtain

$$\begin{aligned} g_{(8,3)}(6, P(1)) &= 8 + \frac{8\log_2 6 - 24}{3 - \log_2[1 - P(1)]} \\ &> 8 + \frac{8\log_2 6 - 24}{3 - \log_2(1 - 0.5)} \\ &> 7 = L_\nu(6), \end{aligned}$$

for all $P(1) \in (0.5, 1 - 2^{-\frac{19}{7}}]$.

- d) When $a \geq 8$, based on Equation (29), we only need to show that

$$h_{(8,3)}(t, x) - \tilde{\Delta}(t) \geq 2\lfloor \log_2(1+t) \rfloor \quad (36)$$

for all $3 \leq t \in \mathbb{N}$ and $1 < x \leq \frac{19}{7}$. When $3 \leq t \leq 6$, the left-hand side of Equation (36) is

$$\begin{aligned} & h_{(8,3)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{8}{3+x} - 1 \right) \times t + \frac{8x}{3+x} - 1 - 0 \\ &\geq \left(\frac{8}{3+x} - 1 \right) \times 3 + \frac{8x}{3+x} - 1 \\ &= 4 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

When $t = 7$, the left-hand side of Equation (36) is

$$\begin{aligned} & h_{(8,3)}(7, x) - \tilde{\Delta}(7) \\ &= \left(\frac{8}{3+x} - 1 \right) \times 7 + \frac{8x}{3+x} - 1 + 1 \\ &= 1 + \frac{32}{3+x} \geq 1 + \frac{32}{3+\frac{19}{7}} \\ &= 6.6 > 2\lfloor \log_2(1+7) \rfloor, \end{aligned}$$

for all $x \in (1, \frac{19}{7}]$. When $t \geq 8$, we prove that Equation (36) holds by proving that

$$h_{(8,3)}(t, x) \geq 2\lfloor \log_2(1+t) \rfloor \quad (37)$$

holds when $8 \leq t \in \mathbb{N}$ and $1 < x \leq \frac{19}{7}$. When $t = 8$, the left-hand side of Equation (37) is

$$\begin{aligned} h_{(8,3)}(8, x) &= \left(\frac{8}{3+x} - 1 \right) \times 8 + \frac{8x}{3+x} - 1 \\ &= 2 + \frac{40}{3+x} - 1 \\ &\geq \frac{40}{3+\frac{19}{7}} - 1 \\ &= 6 = 2\lfloor \log_2(1+8) \rfloor. \end{aligned}$$

Thereafter, each increase of 2 on the right-hand side of Equation (37) increases t by at least $15 - 8 = 7$, and thus the left-hand side of Equation (37) increases by at least

$$\left(\frac{8}{3+x} - 1 \right) \times 7 \geq \left(\frac{8}{3+\frac{19}{7}} - 1 \right) \times 7 = 2.8.$$

Therefore, Equation (37) holds when $8 \leq t \in \mathbb{N}$ and $1 < x \leq \frac{19}{7}$.

- 4) When $3 \leq a \leq 7$, we prove that $L_\nu(a) \leq g_{(15,8)}(a, P(1))$ holds using the same approach as that employed in 3). When $8 \leq a \in \mathcal{A}$, based on Equation (29), we only need to show that

$$h_{(15,8)}(t, x) - \tilde{\Delta}(t) \geq 2\lfloor \log_2(1+t) \rfloor \quad (38)$$

for all $3 \leq t \in \mathbb{N}$ and $\frac{19}{7} < x \leq \frac{41}{8}$. We consider the following seven cases.

- a) When $3 \leq t \leq 6$, the left-hand side of Equation (38) is

$$\begin{aligned} & h_{(15,8)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{15}{8+x} - 1 \right) \times t + \frac{15x}{8+x} - 1 - 0 \\ &\geq \left(\frac{15}{8+x} - 1 \right) \times 3 + \frac{15x}{8+x} - 1 \\ &= 11 - \frac{75}{8+x} > 11 - \frac{75}{8+\frac{19}{7}} \\ &= 4 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- b) When $t = 7$, the left-hand side of Equation (38) is

$$\begin{aligned} & h_{(15,8)}(7, x) - \tilde{\Delta}(7) \\ &= \left(\frac{15}{8+x} - 1 \right) \times 7 + \frac{15x}{8+x} - 1 + 1 \\ &= 8 - \frac{15}{8+x} > 8 - \frac{15}{8+\frac{19}{7}} \\ &= 6.6 > 2\lfloor \log_2(1+7) \rfloor. \end{aligned}$$

- c) When $8 \leq t \leq 14$, the left-hand side of Equation (38) is

$$\begin{aligned} & h_{(15,8)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{15}{8+x} - 1 \right) \times t + \frac{15x}{8+x} - 1 - 0 \\ &\geq \left(\frac{15}{8+x} - 1 \right) \times 8 + \frac{15x}{8+x} - 1 \\ &= 6 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- d) When $15 \leq t \leq 24$, the left-hand side of Equation (38) is

$$\begin{aligned} & h_{(15,8)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{15}{8+x} - 1 \right) \times t + \frac{15x}{8+x} - 1 + 1 \\ &\geq \left(\frac{15}{8+x} - 1 \right) \times 15 + \frac{15x}{8+x} \\ &= \frac{105}{8+x} \geq \frac{105}{8+\frac{41}{8}} \\ &= 8 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- e) When $25 \leq t \leq 30$, the left-hand side of Equation (38) is

$$\begin{aligned} & h_{(15,8)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{15}{8+x} - 1 \right) \times t + \frac{15x}{8+x} - 1 + 0 \\ &\geq \left(\frac{15}{8+x} - 1 \right) \times 25 + \frac{15x}{8+x} - 1 \\ &= \frac{255}{8+x} - 11 \geq \frac{255}{8+\frac{41}{8}} - 11 \\ &> 8 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- f) When $31 \leq t \leq 40$, the left-hand side of Equation (38) is

$$\begin{aligned} & h_{(15,8)}(t, x) - \tilde{\Delta}(t) \\ & \geq \left(\frac{15}{8+x} - 1 \right) \times t + \frac{15x}{8+x} - 1 + 1 \\ & \geq \left(\frac{15}{8+x} - 1 \right) \times 31 + \frac{15x}{8+x} \\ & = \frac{345}{8+x} - 16 \geq \frac{345}{8+\frac{41}{8}} - 16 \\ & > 10 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- g) When $t \geq 41$, we prove that Equation (38) holds by proving that

$$h_{(15,8)}(t, x) \geq 2\lfloor \log_2(1+t) \rfloor \quad (39)$$

holds when $41 \leq t \in \mathbb{N}$ and $\frac{19}{7} < x \leq \frac{41}{8}$. When $t = 41$, the left-hand side of Equation (39) is

$$\begin{aligned} h_{(15,8)}(41, x) &= \left(\frac{15}{8+x} - 1 \right) \times 41 + \frac{15x}{8+x} - 1 \\ &= \frac{495}{8+x} - 27 \geq \frac{495}{8+\frac{41}{8}} - 27 \\ &> 10 = 2\lfloor \log_2(1+41) \rfloor. \end{aligned}$$

Thereafter, each increase of 2 on the right-hand side of Equation (39) increases t by at least $63 - 41 = 22$, and thus, the left-hand side of Equation (39) increases by at least

$$\left(\frac{15}{8+x} - 1 \right) \times 22 \geq \left(\frac{15}{8+\frac{41}{8}} - 1 \right) \times 22 = \frac{22}{7}.$$

Therefore, Equation (39) holds when $41 \leq t \in \mathbb{N}$ and $\frac{19}{7} < x \leq \frac{41}{8}$.

- 5) When $3 \leq a \leq 7$, we prove that $L_\nu(a) \leq g_{(23,15)}(a, P(1))$ holds using the same approach as that employed for 3). When $8 \leq a \in \mathcal{A}$, from Equation (29), we only need to show that

$$h_{(23,15)}(t, x) - \tilde{\Delta}(t) \geq 2\lfloor \log_2(1+t) \rfloor \quad (40)$$

for all $3 \leq t \in \mathbb{N}$ and $\frac{41}{8} < x \leq \frac{65}{11}$. We consider the following ten cases.

- a) When $3 \leq t \leq 6$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{23}{15+x} - 1 \right) \times t + \frac{23x}{15+x} - 1 - 0 \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 3 + \frac{23x}{15+x} - 1 \\ &= 19 - \frac{276}{15+x} > 19 - \frac{276}{15+\frac{41}{8}} \\ &> 5 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- b) When $t = 7$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(7, x) - \tilde{\Delta}(7) \\ &= \left(\frac{23}{15+x} - 1 \right) \times 7 + \frac{23x}{15+x} - 1 + 1 \\ &= 16 - \frac{184}{15+x} > 16 - \frac{184}{15+\frac{41}{8}} \\ &= \frac{48}{7} > 2\lfloor \log_2(1+7) \rfloor. \end{aligned}$$

- c) When $8 \leq t \leq 14$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 8 + \frac{23x}{15+x} - 1 \\ &= 14 - \frac{161}{15+x} > 14 - \frac{161}{15+\frac{41}{8}} \\ &= 6 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- d) When $15 \leq t \leq 24$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{23}{15+x} - 1 \right) \times t + \frac{23x}{15+x} - 1 + 1 \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 15 + \frac{23x}{15+x} \\ &= 8 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- e) When $25 \leq t \leq 30$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 25 + \frac{23x}{15+x} - 1 \\ &= \frac{230}{15+x} - 3 \geq \frac{230}{15+\frac{65}{11}} - 3 \\ &= 8 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- f) When $31 \leq t \leq 36$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{23}{15+x} - 1 \right) \times t + \frac{23x}{15+x} - 1 + 2 \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 31 + \frac{23x}{15+x} + 1 \\ &= \frac{368}{15+x} - 7 \geq \frac{368}{15+\frac{65}{11}} - 7 \\ &= 10.6 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- g) When $37 \leq t \leq 50$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{23}{15+x} - 1 \right) \times t + \frac{23x}{15+x} - 1 + 1 \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 37 + \frac{23x}{15+x} \\ &= \frac{506}{15+x} - 14 \geq \frac{506}{15 + \frac{65}{11}} - 14 \\ &= 10.2 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- h) When $51 \leq t \leq 62$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{23}{15+x} - 1 \right) \times t + \frac{23x}{15+x} - 1 \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 51 + \frac{23x}{15+x} - 1 \\ &= \frac{828}{15+x} - 29 \geq \frac{828}{15 + \frac{65}{11}} - 29 \\ &= 10.6 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- i) When $63 \leq t \leq 84$, the left-hand side of Equation (40) is

$$\begin{aligned} & h_{(23,15)}(t, x) - \tilde{\Delta}(t) \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times t + \frac{23x}{15+x} - 1 + 1 \\ &\geq \left(\frac{23}{15+x} - 1 \right) \times 63 + \frac{23x}{15+x} \\ &= \frac{1104}{15+x} - 40 \geq \frac{1104}{15 + \frac{65}{11}} - 40 \\ &= 12.8 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- j) When $t \geq 85$, we prove that Equation (40) holds by proving that

$$h_{(23,15)}(t, x) \geq 2\lfloor \log_2(1+t) \rfloor \quad (41)$$

holds when $85 \leq t \in \mathbb{N}$ and $\frac{41}{8} < x \leq \frac{65}{11}$. When $t = 85$, the left-hand side of Equation (41) is

$$\begin{aligned} h_{(23,15)}(85, x) &= \left(\frac{23}{15+x} - 1 \right) \times 85 + \frac{23x}{15+x} - 1 \\ &= \frac{1610}{15+x} - 63 \geq \frac{1610}{15 + \frac{65}{11}} - 63 \\ &= 14 > 2\lfloor \log_2(1+85) \rfloor. \end{aligned}$$

Thereafter, each increase of 2 on the right-hand side of Equation (41) increases t by at least $127 - 85 = 42$, and thus, the left-hand side of Equation (41) increases by at least

$$\left(\frac{23}{15+x} - 1 \right) \times 42 \geq \left(\frac{23}{15 + \frac{65}{11}} - 1 \right) \times 42 = 4.2.$$

Therefore, Equation (41) holds when $85 \leq t \in \mathbb{N}$ and $\frac{41}{8} < x \leq \frac{65}{11}$.

- 6) When $3 \leq a \leq 7$, we prove that $L_\nu(a) \leq g_{(34,25)}(a, P(1))$ holds using the same approach as that employed for 3). When $8 \leq a \in \mathcal{A}$, based on Equation (29), we only need to show that

$$h_{(34,25)}(t, x) - \tilde{\Delta}(t) \geq 2\lfloor \log_2(1+t) \rfloor \quad (42)$$

for all $3 \leq t \in \mathbb{N}$ and $\frac{65}{11} < x \leq \frac{83}{13}$. We consider the following nine cases.

- a) When $3 \leq t \leq 6$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{34}{25+x} - 1 \right) \times t + \frac{34x}{25+x} - 1 \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 3 + \frac{34x}{25+x} - 1 \\ &= 30 - \frac{748}{25+x} > 30 - \frac{748}{25 + \frac{65}{11}} \\ &= 5.8 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- b) When $7 \leq t \leq 14$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times t + \frac{34x}{25+x} - 1 \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 7 + \frac{34x}{25+x} - 1 \\ &= 26 - \frac{612}{25+x} > 26 - \frac{612}{25 + \frac{65}{11}} \\ &= 6.2 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- c) When $15 \leq t \leq 24$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{34}{25+x} - 1 \right) \times t + \frac{34x}{25+x} - 1 + 1 \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 15 + \frac{34x}{25+x} \\ &= 19 - \frac{340}{25+x} > 19 - \frac{340}{25 + \frac{65}{11}} \\ &= 8 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- d) When $25 \leq t \leq 30$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{34}{25+x} - 1 \right) \times t + \frac{34x}{25+x} - 1 \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 25 + \frac{34x}{25+x} - 1 \\ &= 8 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- e) When $31 \leq t \leq 36$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{34}{25+x} - 1 \right) \times t + \frac{34x}{25+x} - 1 + 2 \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 31 + \frac{34x}{25+x} + 1 \\ &= \frac{204}{25+x} + 4 \geq \frac{204}{25+\frac{83}{13}} + 4 \\ &= 10.5 > 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- f) When $37 \leq t \leq 50$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &= \left(\frac{34}{25+x} - 1 \right) \times t + \frac{34x}{25+x} - 1 + 1 \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 37 + \frac{34x}{25+x} \\ &= \frac{408}{25+x} - 3 \geq \frac{408}{25+\frac{83}{13}} - 3 \\ &= 10 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- g) When $51 \leq t \leq 62$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 51 + \frac{34x}{25+x} - 1 \\ &= \frac{884}{25+x} - 18 \geq \frac{884}{25+\frac{83}{13}} - 18 \\ &> 10 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- h) When $63 \leq t \leq 84$, the left-hand side of Equation (42) is

$$\begin{aligned} & h_{(34,25)}(t, x) - \tilde{\Delta}(t) \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times t + \frac{34x}{25+x} - 1 + 1 \\ &\geq \left(\frac{34}{25+x} - 1 \right) \times 63 + \frac{34x}{25+x} \\ &= \frac{1292}{25+x} - 29 \geq \frac{1292}{25+\frac{83}{13}} - 29 \\ &> 12 = 2\lfloor \log_2(1+t) \rfloor. \end{aligned}$$

- i) When $t \geq 85$, we prove that Equation (42) holds by proving that

$$h_{(34,25)}(t, x) \geq 2\lfloor \log_2(1+t) \rfloor \quad (43)$$

holds when $85 \leq t \in \mathbb{N}$ and $\frac{65}{11} < x \leq \frac{83}{13}$. When $t = 85$, the left-hand side of Equation (43) is

$$\begin{aligned} h_{(34,25)}(85, x) &= \left(\frac{34}{25+x} - 1 \right) \times 85 + \frac{34x}{25+x} - 1 \\ &= \frac{2040}{25+x} - 52 \geq \frac{2040}{25+\frac{83}{13}} - 52 \\ &= 13 > 2\lfloor \log_2(1+85) \rfloor. \end{aligned}$$

Thereafter, each increase of 2 on the right-hand side of Equation (43) increases t by at least $127 - 85 = 42$, and thus, the left-hand side of Equation (43) increases by at least

$$\left(\frac{34}{25+x} - 1 \right) \times 42 \geq \left(\frac{34}{25+\frac{83}{13}} - 1 \right) \times 42 = 3.5.$$

Therefore, Equation (43) holds when $85 \leq t \in \mathbb{N}$ and $\frac{65}{11} < x \leq \frac{83}{13}$.

- 7) When $3 \leq a \leq 7$, we prove that $L_\nu(a) \leq g_{(47,37)}(a, P(1))$ holds using the same way as that employed for 3). When $8 \leq a \in \mathcal{A}$, we prove that $L_\nu(a) \leq g_{(47,37)}(a, P(1))$ holds using the same approach as that employed for 5) and 6).
 8) When $3 \leq a \leq 7$, we prove that $L_\nu(a) \leq g_{(62,51)}(a, P(1))$ holds using the same approach as that employed for 3). When $8 \leq a \in \mathcal{A}$, we prove that $L_\nu(a) \leq g_{(62,51)}(a, P(1))$ holds using the same approach as that employed for 5) and 6).
 9) When $3 \leq a \leq 7$, we prove that $L_\nu(a) \leq g_{(98,85)}(a, P(1))$ holds using the same approach as that employed for 3). When $8 \leq a \in \mathcal{A}$, we prove that $L_\nu(a) \leq g_{(98,85)}(a, P(1))$ holds using the same approach as that employed for 5) and 6).
 10) When $3 \leq a \leq 7$, we prove that $L_\nu(a) \leq g_{(142,127)}(a, P(1))$ holds using the same approach as that employed for 3). When $8 \leq a \in \mathcal{A}$, we prove that $L_\nu(a) \leq g_{(142,127)}(a, P(1))$ holds using the same approach as that employed for 5) and 6). $\square$

APPENDIX B

PROOF OF THE REMAINING PART OF THEOREM 5

Proof. 4) Case $1 - 2^{-\frac{19}{7}} < P(1) \leq 0.92$: From 4) of Lemma 6, we know that $c_1 = 15$, and $c_2 = 8$. Consider $L = 4$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ &\leq Q_{(15,8)}(P(1), P(2), 4) + D_{(15,8)}(P(1)) \\ &\stackrel{(a)}{\leq} Q_{(15,8)}(P(1), 1 - P(1), 4) + D_{(15,8)}(P(1)), \end{aligned}$$

where (a) is true because

$$\begin{aligned} R_{(c_1, c_2)}(P(1), L) &= 3 + D_{(15,8)}(P(1)) \left(\log_2(1 - P(1)) \right. \\ &\quad \left. + \sum_{n=2}^3 \log_2 C_n - 3 \log_2 C_4 \right) > 0 \end{aligned}$$

for all $P(1) \in (1 - 2^{-\frac{19}{7}}, 0.92]$. Let $f_5(x) \triangleq Q_{(15,8)}(x, 1 - x, 4) + D_{(15,8)}(x)$. By calculating the derivative, we know that $f_5(x)$ is decreasing and then increasing over the interval $(1 - 2^{-\frac{19}{7}}, 0.92]$. Therefore, we obtain

$$\begin{aligned} \frac{A_P(L_\nu)}{\max\{1, H(P)\}} &\leq f_5(P(1)) \\ &\leq \max\{f_5(1 - 2^{-\frac{19}{7}}), f_5(0.92)\} \\ &= f_5(0.92) < 1.9999 \end{aligned}$$

for all $P(1) \in (1 - 2^{-\frac{19}{7}}, 0.92]$.

- 5) Case $0.92 < P(1) \leq 1 - 2^{-\frac{41}{8}} \approx 0.97134$: From 4) of Lemma 6, we know that $c_1 = 15$, and $c_2 = 8$. If $L = 5$, then

$$R_{(c_1, c_2)}(P(1), L) = 3 + D_{(15, 8)}(P(1)) \left(\log_2(1 - P(1)) + \sum_{n=2}^4 \log_2 C_n - 4 \log_2 C_5 \right).$$

Let

$$j_2(x) \triangleq 3 + D_{(15, 8)}(x) \left(\log_2(1 - x) + \sum_{n=2}^4 \log_2 C_n - 4 \log_2 C_5 \right).$$

By calculating the derivative, we know that $j_2(x)$ is strictly decreasing over the interval $(0.92, 1 - 2^{-\frac{41}{8}}]$, and the unique zero point over the interval is $x_2 \approx 0.95602$.

- a) When $0.92 < P(1) \leq x_2$, owing to Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(15, 8)}(P(1), P(2), 5) + D_{(15, 8)}(P(1)) \\ & \stackrel{(a)}{\leq} Q_{(15, 8)}(P(1), 1 - P(1), 5) + D_{(15, 8)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(15, 8)}(P(1), 5) = j_2(P(1)) \geq 0$ for all $P(1) \in (0.92, x_2]$. Let $f_6(x) \triangleq Q_{(15, 8)}(x, 1 - x, 5) + D_{(15, 8)}(x)$. By calculating the derivative, we know that $f_6(x)$ is strictly decreasing over the interval $(0.92, x_2]$. Thus, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_6(P(1)) < f_6(0.92) < 2.0313$$

for all $P(1) \in (0.92, x_2]$.

- b) When $x_2 < P(1) \leq 1 - 2^{-\frac{41}{8}}$, according to Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(15, 8)}(P(1), P(2), 5) + D_{(15, 8)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(15, 8)}(P(1), 5) + D_{(15, 8)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(15, 8)}(P(1), 5) = j_2(P(1)) < 0$ for all $P(1) \in (x_2, 1 - 2^{-\frac{41}{8}}]$. Let $f_7(x) \triangleq J_{(15, 8)}(x, 5) + D_{(15, 8)}(x)$. By calculating the derivative, we know that $f_7(x)$ is increasing and then decreasing over the interval $(x_2, 1 - 2^{-\frac{41}{8}}]$, and $f_7(x)$ takes its maximum value over $(x_2, 1 - 2^{-\frac{41}{8}}]$ at $x_3 \approx 0.96883$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_7(P(1)) \leq f_7(x_3) < 2.0345$$

for all $P(1) \in (x_2, 1 - 2^{-\frac{41}{8}}]$.

- 6) Case $1 - 2^{-\frac{41}{8}} < P(1) \leq 1 - 2^{-\frac{65}{11}} \approx 0.98336$: From 5) of Lemma 6, we know that $c_1 = 23$, and $c_2 = 15$. Consider $L = 6$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(23, 15)}(P(1), P(2), 6) + D_{(23, 15)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(23, 15)}(P(1), 6) + D_{(23, 15)}(P(1)), \end{aligned}$$

where (a) is true because

$$R_{(c_1, c_2)}(P(1), L) = 3 + D_{(23, 15)}(P(1)) \left(\log_2(1 - P(1)) + \sum_{n=2}^5 \log_2 C_n - 5 \log_2 C_6 \right) < 0$$

for all $P(1) \in (1 - 2^{-\frac{41}{8}}, 1 - 2^{-\frac{65}{11}}]$. Let $f_8(x) \triangleq J_{(23, 15)}(x, 6) + D_{(23, 15)}(x)$. By calculating the derivative, we know that $f_8(x)$ is increasing and then decreasing over the interval $(1 - 2^{-\frac{41}{8}}, 1 - 2^{-\frac{65}{11}}]$, and $f_8(x)$ takes its maximum value over $(1 - 2^{-\frac{41}{8}}, 1 - 2^{-\frac{65}{11}}]$ at $x_4 \approx 0.98053$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_8(P(1)) \leq f_8(x_4) < 2.0380$$

for all $P(1) \in (1 - 2^{-\frac{41}{8}}, 1 - 2^{-\frac{65}{11}}]$.

- 7) Case $1 - 2^{-\frac{65}{11}} < P(1) \leq 1 - 2^{-\frac{83}{13}} \approx 0.98803$: From 6) of Lemma 6, we know that $c_1 = 34$, and $c_2 = 25$. Consider $L = 8$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(34, 25)}(P(1), P(2), 8) + D_{(34, 25)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(34, 25)}(P(1), 8) + D_{(34, 25)}(P(1)), \end{aligned}$$

where (a) is true because

$$R_{(c_1, c_2)}(P(1), L) = 3 + D_{(34, 25)}(P(1)) \left(\log_2(1 - P(1)) + \sum_{n=2}^7 \log_2 C_n - 7 \log_2 C_8 \right) < 0$$

for all $P(1) \in (1 - 2^{-\frac{65}{11}}, 1 - 2^{-\frac{83}{13}}]$. Let $f_9(x) \triangleq J_{(34, 25)}(x, 8) + D_{(34, 25)}(x)$. By calculating the derivative, we know that $f_9(x)$ is increasing and then decreasing over the interval $(1 - 2^{-\frac{65}{11}}, 1 - 2^{-\frac{83}{13}}]$, and $f_9(x)$ takes its maximum value over $(1 - 2^{-\frac{65}{11}}, 1 - 2^{-\frac{83}{13}}]$ at $x_5 \approx 0.98735$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_9(P(1)) \leq f_9(x_5) < 2.0376$$

for all $P(1) \in (1 - 2^{-\frac{65}{11}}, 1 - 2^{-\frac{83}{13}}]$.

- 8) Case $1 - 2^{-\frac{83}{13}} < P(1) \leq 1 - 2^{-\frac{103}{15}} \approx 0.99143$: From 7) of Lemma 6, we know that $c_1 = 47$, and $c_2 = 37$. If $L = 12$, then

$$R_{(c_1, c_2)}(P(1), L) = 3 + D_{(47, 37)}(P(1)) \left(\log_2(1 - P(1)) + \sum_{n=2}^{11} \log_2 C_n - 11 \log_2 C_{12} \right).$$

Let

$$j_3(x) \triangleq 3 + D_{(47,37)}(x) \left(\log_2(1-x) + \sum_{n=2}^{11} \log_2 C_n - 11 \log_2 C_{12} \right).$$

By calculating the derivative, we know that $j_3(x)$ is strictly decreasing over the interval $(1 - 2^{-\frac{83}{13}}, 1 - 2^{-\frac{103}{15}}]$, and the unique zero point over this interval is $x_6 \approx 0.98876$.

- a) When $1 - 2^{-\frac{83}{13}} < P(1) \leq x_6$, based on Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(47,37)}(P(1), P(2), 12) + D_{(47,37)}(P(1)) \\ & \stackrel{(a)}{\leq} Q_{(47,37)}(P(1), 1 - P(1), 12) + D_{(47,37)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(47,37)}(P(1), 12) = j_3(P(1)) \geq 0$ for all $P(1) \in (1 - 2^{-\frac{83}{13}}, x_6]$. Let $f_{10}(x) \triangleq Q_{(47,37)}(x, 1 - x, 12) + D_{(47,37)}(x)$. By calculating the derivative, we know that $f_{10}(x)$ is strictly decreasing over the interval $(1 - 2^{-\frac{83}{13}}, x_6]$. Thus, we obtain

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{10}(P(1)) \\ & < f_{10}(1 - 2^{-\frac{83}{13}}) < 2.0375 \end{aligned}$$

for all $P(1) \in (1 - 2^{-\frac{83}{13}}, x_6]$.

- b) When $x_6 < P(1) \leq 1 - 2^{-\frac{103}{15}}$, owing to Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(47,37)}(P(1), P(2), 12) + D_{(47,37)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(47,37)}(P(1), 12) + D_{(47,37)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(47,37)}(P(1), 12) = j_3(P(1)) < 0$ for all $P(1) \in (x_6, 1 - 2^{-\frac{103}{15}}]$. Let $f_{11}(x) \triangleq J_{(47,37)}(x, 12) + D_{(47,37)}(x)$. By calculating the derivative, we know that $f_{11}(x)$ is increasing and then decreasing over the interval $(x_6, 1 - 2^{-\frac{103}{15}}]$, and $f_{11}(x)$ takes its maximum value over $(x_6, 1 - 2^{-\frac{103}{15}}]$ at $x_7 \approx 0.99114$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{11}(P(1)) \leq f_{11}(x_7) < 2.0381$$

for all $P(1) \in (x_6, 1 - 2^{-\frac{103}{15}}]$.

- 9) Case $1 - 2^{-\frac{103}{15}} < P(1) \leq 0.9927$: From 8) of Lemma 6, we know that $c_1 = 62$, and $c_2 = 51$. If $L = 15$, then

$$\begin{aligned} R_{(c_1, c_2)}(P(1), L) &= 3 + D_{(62,51)}(P(1)) \left(\log_2(1 - P(1)) \right. \\ & \quad \left. + \sum_{n=2}^{14} \log_2 C_n - 14 \log_2 C_{15} \right). \end{aligned}$$

Let

$$j_4(x) \triangleq 3 + D_{(62,51)}(x) \left(\log_2(1-x) + \sum_{n=2}^{14} \log_2 C_n - 14 \log_2 C_{15} \right).$$

By calculating the derivative, we know that $j_4(x)$ is strictly decreasing over the interval $(1 - 2^{-\frac{103}{15}}, 0.9927]$, and the unique zero point over the interval is $x_8 \approx 0.99195$.

- a) When $1 - 2^{-\frac{103}{15}} < P(1) \leq x_8$, due to Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(62,51)}(P(1), P(2), 15) + D_{(62,51)}(P(1)) \\ & \stackrel{(a)}{\leq} Q_{(62,51)}(P(1), 1 - P(1), 15) + D_{(62,51)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(62,51)}(P(1), 15) = j_4(P(1)) \geq 0$ for all $P(1) \in (1 - 2^{-\frac{103}{15}}, x_8]$. Let $f_{12}(x) \triangleq Q_{(62,51)}(x, 1 - x, 15) + D_{(62,51)}(x)$. By calculating the derivative, we know that $f_{12}(x)$ is strictly decreasing over the interval $(1 - 2^{-\frac{103}{15}}, x_8]$. Thus, we obtain

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{12}(P(1)) \\ & < f_{12}(1 - 2^{-\frac{103}{15}}) < 2.0386 \end{aligned}$$

for all $P(1) \in (1 - 2^{-\frac{103}{15}}, x_8]$.

- b) When $x_8 < P(1) \leq 0.9927$, because of Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(62,51)}(P(1), P(2), 15) + D_{(62,51)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(62,51)}(P(1), 15) + D_{(62,51)}(P(1)), \end{aligned}$$

where (a) is true because $R_{(62,51)}(P(1), 15) = j_4(P(1)) < 0$ for all $P(1) \in (x_8, 0.9927]$. Let $f_{13}(x) \triangleq J_{(62,51)}(x, 15) + D_{(62,51)}(x)$. By calculating the derivative, we know that $f_{13}(x)$ is strictly increasing over the interval $(x_8, 0.9927]$. Therefore, we obtain

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{13}(P(1)) \\ & \leq f_{13}(0.9927) < 2.0386 \end{aligned}$$

for all $P(1) \in (x_8, 0.9927]$.

- 10) Case $0.9927 < P(1) \leq 1 - 2^{-\frac{68}{9}} \approx 0.99468$: From 8) of Lemma 6, we know that $c_1 = 62$, and $c_2 = 51$. Consider $L = 16$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(62,51)}(P(1), P(2), 16) + D_{(62,51)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(62,51)}(P(1), 16) + D_{(62,51)}(P(1)), \end{aligned}$$

where (a) is true because

$$\begin{aligned} R_{(c_1, c_2)}(P(1), L) &= 3 + D_{(62,51)}(P(1)) \left(\log_2(1 - P(1)) \right. \\ & \quad \left. + \sum_{n=2}^{15} \log_2 C_n - 15 \log_2 C_{16} \right) < 0 \end{aligned}$$

for all $P(1) \in (0.9927, 1 - 2^{-\frac{68}{9}}]$. Let $f_{14}(x) \triangleq J_{(62,51)}(x, 16) + D_{(62,51)}(x)$. By calculating the derivative, we know that $f_{14}(x)$ is increasing and then decreasing over the interval $(0.9927, 1 - 2^{-\frac{68}{9}}]$, and $f_{14}(x)$ takes its maximum value over $(0.9927, 1 - 2^{-\frac{68}{9}}]$ at $x_9 \approx 0.99339$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{14}(P(1)) \leq f_{14}(x_9) < 2.0386$$

for all $P(1) \in (0.9927, 1 - 2^{-\frac{68}{9}}]$.

- 11) Case $1 - 2^{-\frac{68}{9}} < P(1) \leq 1 - 2^{-\frac{94}{11}} \approx 0.99732$: From 9) of Lemma 6, we know that $c_1 = 98$, and $c_2 = 85$. Consider $L = 18$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(98,85)}(P(1), P(2), 18) + D_{(98,85)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(98,85)}(P(1), 18) + D_{(98,85)}(P(1)), \end{aligned}$$

where (a) is true because

$$\begin{aligned} R_{(c_1, c_2)}(P(1), L) &= 3 + D_{(98,85)}(P(1)) \left(\log_2(1 - P(1)) \right. \\ & \quad \left. + \sum_{n=2}^{17} \log_2 C_n - 17 \log_2 C_{18} \right) < 0 \end{aligned}$$

for all $P(1) \in (1 - 2^{-\frac{68}{9}}, 1 - 2^{-\frac{94}{11}}]$. Let $f_{15}(x) \triangleq J_{(98,85)}(x, 18) + D_{(98,85)}(x)$. By calculating the derivative, we know that $f_{15}(x)$ is increasing and then decreasing over the interval $(1 - 2^{-\frac{68}{9}}, 1 - 2^{-\frac{94}{11}}]$, and $f_{15}(x)$ takes its maximum value over $(1 - 2^{-\frac{68}{9}}, 1 - 2^{-\frac{94}{11}}]$ at $x_{10} \approx 0.99586$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{15}(P(1)) \leq f_{15}(x_{10}) < 2.0386$$

for all $P(1) \in (1 - 2^{-\frac{68}{9}}, 1 - 2^{-\frac{94}{11}}]$.

- 12) Case $1 - 2^{-\frac{94}{11}} < P(1) \leq 1 - 2^{-\frac{833}{65}} \approx 0.99986$: From 10) of Lemma 6, we know that $c_1 = 142$, and $c_2 = 127$. Consider $L = 18$. From Equation (33), it follows that

$$\begin{aligned} & \frac{A_P(L_\nu)}{\max\{1, H(P)\}} \\ & \leq Q_{(142,127)}(P(1), P(2), 18) + D_{(142,127)}(P(1)) \\ & \stackrel{(a)}{\leq} J_{(142,127)}(P(1), 18) + D_{(142,127)}(P(1)), \end{aligned}$$

where (a) is true because

$$\begin{aligned} R_{(c_1, c_2)}(P(1), L) &= 3 + D_{(142,127)}(P(1)) \left(\log_2(1 - P(1)) \right. \\ & \quad \left. + \sum_{n=2}^{17} \log_2 C_n - 17 \log_2 C_{18} \right) < 0 \end{aligned}$$

for all $P(1) \in (1 - 2^{-\frac{94}{11}}, 1 - 2^{-\frac{833}{65}}]$. Let $f_{16}(x) \triangleq J_{(142,127)}(x, 18) + D_{(142,127)}(x)$. By calculating the derivative, we know that $f_{16}(x)$ is strictly decreasing over the interval $(1 - 2^{-\frac{94}{11}}, 1 - 2^{-\frac{833}{65}}]$. Therefore, we obtain

$$\frac{A_P(L_\nu)}{\max\{1, H(P)\}} \leq f_{16}(P(1)) < f_{16}(1 - 2^{-\frac{94}{11}}) < 2.0372$$

for all $P(1) \in (1 - 2^{-\frac{94}{11}}, 1 - 2^{-\frac{833}{65}}]$. □

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